STUDENT RESEARCH MONOGRAPH

A Comprehensive Approach to Tail-Risk Estimation via Extreme Value Theory

Forecasting and stress-testing warehouse fulfilment losses in a risk-management setting: a complete mathematical treatment

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Independent student research.

Empirical substrate: UCI Online Retail (Chen, 2015), 1 December 2010 – 9 December 2011.

Companion code: github.com/Shridathj/Warehouse-Anomaly-Detection-Tail-Risk-Stress-Testing

Scope, in one paragraph.

This paper derives, from first principles, every estimator used in the warehouse dragon-order tail-risk pipeline: extreme-value tail indices on order quantity, a peaks-over-threshold generalised Pareto model of daily realised loss, log-normal delay simulation, Monte Carlo value-at-risk and expected shortfall, quantile treatment effects, Hawkes maximum likelihood, a Kalman local-linear-trend state-space forecast, and purged expanding-window backtests with Kupiec and Christoffersen tests. Delays are synthetic. Dragon labels are assigned, not observed. Average treatment effects of the surge indicator are therefore a negative control (they are expected to be near zero). The object of causal interest is the quantile contrast in order value. All sterling figures are directional Monte Carlo outputs under stated assumptions; they are not operational loss forecasts. Where the companion code departs from the intended theory, the departure is recorded in an implementation note.

Abstract

High-value "dragon" orders are rare, heavy-tailed, and operationally expensive when fulfilment service degrades. This paper develops a single mathematical chain from the Fisher–Tippett–Gnedenko theorem to a purged backtest of warehouse tail loss, and instantiates every step on the UCI Online Retail data set under two exposure conventions: gross (maximum booked demand) and customer–SKU netted (demand after cancellations). The diagnostic core is extreme value theory. Order quantities on high-volume SKUs lie in the Fréchet domain: Hill, moment, and GEV block-maxima estimators all return a positive tail index ($\xi \approx 0.47$–0.55 gross; 0.48–0.83 netted) and the mean-excess plot is linearly increasing. That chapter is a domain diagnosis; the only generalised Pareto MLE in the pipeline is fitted later, to daily realised loss. An AMSE curve is computed as a threshold diagnostic and is not used to fit a GPD on quantity. Fulfilment delays are not in the public file; they are simulated from value-biased log-normals calibrated to public WERC/CSCMP summaries. Annual preventable loss is then a Monte Carlo functional of dragon revenue, an empirical breach rate, and a 30% gross margin. Because dragon assignment is a function of order value, the average treatment effect of a surge flag on net revenue is a negative control and is empirically −£2 / −£18 (near zero, as required). The informative contrast is the quantile treatment effect (QTE) of the dragon label on order value. Clustering is modelled by an exponential Hawkes process; the twelve-month count is projected by a Kalman local-linear-trend filter (not a Bayesian structural time series). A purged expanding-window backtest, with window-matched VaR, is reported together with Kupiec and Christoffersen tests. The paper's empirical contribution is a reconciliation identity: the several headline sterling numbers in the repository (Monte Carlo ES95, last-window backtest ES95, GPD daily ES, causal annual impact, Hawkes/Kalman P&L bleed) answer different mathematical questions and are not in contradiction once each functional is written down. The scientific limitation is structural: delays are assigned, GPD and Hawkes are fitted in-sample, and one year of public retail tickets is not warehouse telemetry.

Keywords. Extreme value theory; generalised Pareto distribution; Hill estimator; expected shortfall; Hawkes process; Kalman filter; quantile treatment effect; purged backtest; Kupiec test; warehouse fulfilment risk.

Contents

1. Introduction and problem statement

This paper is independent student research on tail-risk estimation for warehouse fulfilment. It is written as a technical paper a practising data scientist can read end to end: every estimator is derived, every sterling headline is identified as a named functional of a named random variable, and every place the companion code parts company with the theory is recorded in an implementation note. The empirical substrate is a public retail file, not warehouse telemetry. The conclusions are about the mathematics of a well-specified pipeline, not about a live distribution centre.

1.1 The operational question

A warehouse does not fail on the mean order. It fails on the right tail: a small number of tickets that are simultaneously large in value and slow to fulfil. In the language of this project those tickets are *dragons*—not a zoological claim, but a name for a rare, value-biased sample (about 3.2 basis points of tickets in the gross scenario, 2.5 after netting) that the pipeline treats as the expensive fulfilment tail. Those tickets dominate holding cost, service-level breaches, and the preventable piece of gross-margin loss. The managerial question is narrow and quantitative. If service on those tickets degrades from a 99th-percentile service level to a 95th-percentile service level, how large is the preventable loss, in sterling, over a one-year horizon, and which interventions move that number?

The question is a tail-risk question, not a forecasting-the-mean question. Sample means, ordinary least squares, and Gaussian value-at-risk are the wrong instruments: the quantity distribution on this file has Pearson kurtosis on the order of 10^5, a coefficient of variation above 13 in the gross scenario, and a maximum ticket of £168,469.60 against a median of £11.80. Extreme value theory exists precisely for this regime. The rest of the pipeline exists to translate a tail index into a loss functional that a risk manager can read: a Monte Carlo distribution of annual preventable loss, a quantile contrast that says how much more valuable dragon tickets are, a point-process forecast of how dragons cluster, a state-space projection of next year's count, and a backtest that asks whether a VaR number is calibrated.

1.2 What is observed, and what is assigned

The only public file with customer, SKU, quantity, price and timestamp at this scale is the UCI Online Retail data set (Chen, 2015): a UK giftware e-tailer, 1 December 2010 to 9 December 2011. It contains *no* warehouse timestamps, no pick-path, no SLA clock, and no dragon label. Every delay in this project is therefore a random variable drawn by the analyst, and every dragon label is a weighted sample from the tickets themselves. That fact is not a footnote; it is the identification structure of the paper. It has two immediate consequences.

First, the average treatment effect of the surge indicator on net revenue is a negative control. Surge is assigned independently of covariates once dragons have been removed, and delay given tier is an independent log-normal. There is no causal path from the surge flag to order value, and holding cost is on the order of 100 basis points of value (0.85% of gross in Scenario 1; the surge-versus-normal delay increment is about 110 bp of ticket value). The ATE must be near zero; empirically it is −£2 (gross) and −£18 (netted). Reporting that number as a policy effect would be a mistake. Second, the quantile treatment effect of the dragon label on order value is partly mechanical: dragons are sampled with probability proportional to value$^{\gamma}$. The QTE is nonetheless the right *descriptive* object for a risk manager—it answers "how much more valuable is a ticket in the dragon tier, at quantile q?"—provided it is not sold as an estimated effect of a treatment that nature applied.

1.3 Two scenarios, on purpose

Scenario 1 keeps every positive-quantity ticket after dropping returns. It is a *gross exposure* view: the warehouse must still provision inventory, labour and working capital against booked demand, including tickets that will later be cancelled. Scenario 2 nets quantity at the (CustomerID, SKU) grain over the whole sample and retains only keys with positive net quantity. It is a *fulfilled-demand* view. The famous 80,995-unit "PAPER CRAFT, LITTLE BIRDIE" ticket (StockCode 23843, £168,469.60) is followed within minutes by a matching negative quantity; it dominates Scenario 1 and disappears from Scenario 2. Both views are economically real. Gross exposure is what a WMS sees at order-entry; net demand is what a P&L eventually recognises. A data governance chapter returns to this point.

1.4 Contributions

- **A complete derivation** of every estimator in the pipeline, from the regular-variation representation of a Fréchet tail through Hill, Dekkers–Einmahl–de Haan, GEV, Pickands–Balkema–de Haan, GPD MLE, AMSE threshold choice, log-normal method of moments, VaR and expected shortfall, Rosenbaum–Rubin propensity scores, Koenker–Bassett quantile regression, Doksum/QTE contrasts, Ogata's Hawkes likelihood, the Kalman filter for a local linear trend, and the Kupiec and Christoffersen coverage tests.
- **An explicit identification statement** for the causal chapter: ATE of surge is a negative control; QTE of dragon on value is the object of interest and is partly by construction.
- **A reconciliation identity** showing that the several sterling headlines in the repository are different functionals of the same simulated file, not competing estimates of one number.
- **Implementation notes** that record, without theatre, where the code departs from the theory: Kalman rather than Bayesian BSTS; an unused seasonal block; Hawkes Ogata recursion that does not decay the current jump (the material reason α_H sits on the 0.99 bound); a broken stationarity guard in Scenario 1 Hawkes; Country and Hour dropped by the loader so propensity covariates collapse; Scenario 1 Hawkes P&L hardcoded; GPD fitted on four daily exceedances in each scenario, with "95%" VaR/ES clamped to the 99th-percentile threshold; Kupiec's p-value returned as 1 when there are zero violations; AMSE Hill computed on excesses with $k = N_u$ and then unused; Scenario 2 PSM/QR outcome equal to order value, not net of holding; Scenario 1 PSM allowing control reuse; Hawkes likelihood truncated at the last event.

1.5 Organisation

Section 2 motivates the method stack. Section 3 writes down the two data maps. Section 4 records the global diagnostics that rule out Gaussian risk. Section 5 is the EVT core. Sections 6–7 turn the tail into a simulated loss and a Monte Carlo ES. Section 8 is the causal chapter, QTE-first. Sections 9–10 forecast counts. Sections 11–12 backtest. Section 13 reconciles money. Sections 14–16 discuss data governance, limits, and what the work does and does not show.

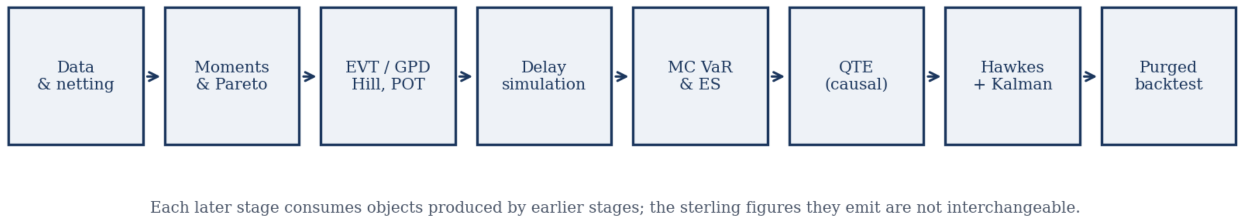

Figure 1. *The pipeline as a directed chain. Later sterling numbers are functionals of earlier objects; they should be read as a family, not as a single "the" loss.*

2. Why these methods, and in this order

2.1 Why not a Gaussian, and why not a black-box detector

Anomaly detection in operations research often means an isolation forest, an autoencoder, or a z-score on a rolling mean. Those tools answer a different question: "which tickets look unlike the others?" The present question is "how heavy is the right tail, and what is the loss functional of that tail under a service-level shock?" A z-score assumes a second moment that is a useful scale. Here the fourth moment is effectively infinite for practical sample sizes (Pearson kurtosis 178,189 on gross quantity). Isolation forests do not produce a tail index, a mean-excess function, or an expected shortfall. Extreme value theory does.

2.2 Extreme value theory, first

EVT is used first because it is the only asymptotically justified model for the far right tail of i.i.d. (or weakly dependent) observations. The Fisher–Tippett–Gnedenko theorem classifies the limit law of maxima. The Pickands–Balkema–de Haan theorem classifies the limit law of excesses over a high threshold. Together they give a one-parameter diagnosis—the sign of ξ—that decides whether the warehouse is in a heavy-tailed world (Fréchet), an exponential world (Gumbel), or a bounded world (Weibull). The quantity chapter uses that diagnosis only: Hill, the moment estimator, GEV block maxima, and a mean-excess slope. A GPD is not fitted to quantity. The only GPD MLE in the repository is the daily-loss fit of Section 11. If $\xi \leq 0$ on quantity, dragons are not a regularly varying phenomenon and the rest of the stack should be rebuilt.

2.3 Simulation, because the SLA clock is missing

Given a tail, one still needs a delay. The file has InvoiceDate, not pick-complete time. The honest move is to simulate delays from a parametric family whose means, coefficients of variation and clip ranges are taken from public WERC DC Measures and CSCMP State of Logistics summaries, and to say so. A log-normal is used because cycle times are positive, right-skewed, and conventionally modelled that way in fulfilment analytics. Dragons are not a random 3.2 bps sample: they are value-biased, because the economic object of interest is the *expensive* tail, not a uniform random ticket that happens to be slow.

2.4 Monte Carlo, because the annual loss is a sum of rare events

Annual preventable loss is a sum, over 365 days, of a random daily dragon revenue times an empirical breach rate times a margin. That sum has no useful closed form once daily revenue is log-normal and the breach rate is a ratio of two sample totals. Monte Carlo is the standard way to read VaR and expected shortfall off a distribution one can sample. The i.i.d. assumption is a computational convenience; clustering is handled in a separate Hawkes chapter and is *not* yet folded into the Monte Carlo paths. LIMITATIONS.md records this deliberately.

2.5 QTE rather than ATE

A mean contrast between surge and non-surge tickets, after propensity matching, estimates the wrong thing: surge is assigned, not discovered, and it does not cause order value. A quantile contrast between dragon and non-dragon order values estimates the object a risk manager actually needs—how the value distribution of the flagged tail sits relative to the rest—at the price of admitting that the flag is itself a function of value. Section 8 writes both estimators down and uses the ATE only as a specification check.

2.6 Hawkes, then a Kalman projection

Dragons are events in time. If they cluster, a Poisson baseline understates short-horizon intensity. A Hawkes process with exponential kernel is the simplest self-exciting model with a closed compensator and a recursive likelihood. A twelve-month *count* forecast, by contrast, is a small time-series problem (thirteen monthly points). A local linear trend Kalman filter is the linear-Gaussian tool for that job. It is not a Bayesian structural time series: there is no posterior over Q and R, no spike-and-slab, no PyMC. The repository's older prose called it BSTS; this paper does not.

2.7 Purged backtests, because leakage is the default

Fitting a tail model on the whole sample and then scoring the whole sample is not a test. An expanding window with a purge gap between train cutoff and test end is the standard way, in quantitative risk, to respect serial dependence and overlapping information (cf. López de Prado's purged cross-validation). Kupiec's unconditional coverage test and Christoffersen's independence test are the textbook pair for VaR exception sequences. With one year of data their power is low; that is stated, not hidden.

3. Data, cleaning, and the two exposure conventions

3.1 The file

Let a raw row be a tuple (InvoiceNo, StockCode, Quantity, InvoiceDate, UnitPrice, CustomerID, Country). The UCI Online Retail file contains 541,909 such rows. CustomerID is missing for a non-trivial share of tickets; those rows are dropped because Scenario 2's netting key requires an identified customer. Unit prices that are non-positive are dropped. After this filter the working sample for Scenario 1 is 397,884 positive-quantity tickets, 3,665 SKUs, spanning 374 calendar days, with gross merchandise value £8,911,407.90.

3.2 Scenario 1: gross booked demand

Write Q_i for quantity and P_i for unit price. Scenario 1 keeps rows with $Q_i > 0$, $P_i > 0$ and CustomerID observed, and defines order value by

$$V_i \;=\; Q_i\, P_i. \tag{1}$$

No cancellation is netted. A subsequent negative-quantity row with the same customer and SKU is simply absent from this file. Gross value is therefore an upper bound on recognised demand and a realistic bound on short-horizon operational exposure: pickers, inventory and working capital are committed when the ticket is booked, not when it survives the returns window.

3.3 Scenario 2: customer–SKU netting

Let the netting key be κ = (CustomerID, SKU), concatenated in the code as a string. Let G be the set of gross picks (Q > 0) and C the set of cancellations (Q < 0), both after the UnitPrice and CustomerID filter. Define

$$N_\kappa = \sum_{i\,:\,\kappa(i)=\kappa} Q_i. \tag{2}$$

Retain keys with $N_\kappa > 0$. For each such key the code keeps a *single* representative row: the gross pick of maximum quantity on that key, and overwrites its quantity and value by the net,

$$Q_\kappa^{\text{net}} = N_\kappa, \qquad V_\kappa^{\text{net}} = N_\kappa\, P_{i^*(\kappa)}, \tag{3}$$

where $i^*(\kappa) = \arg\max \{Q_i : i \text{ in } G, \kappa(i)=\kappa\}$. The resulting table has 265,210 rows (8,905 cancellation rows participated in the netting). This is a strong identifying assumption: all cancellations of a customer–SKU pair, at any time in the year, are charged against that customer's largest pick of that SKU, and the timestamp of the kept row is the timestamp of that largest pick. It is not invoice-level netting, and it is not a twelve-minute matching window. It is an annual, key-level, max-pick representative.

Implementation note — what the repository actually does.
In *src/data/loader.py*, Scenario 1 never drops miscellaneous codes (POST, DOT, BANK CHARGES, gift vouchers). Scenario 2 builds a miscellaneous-code mask, but the compiled pattern is a single concatenated string and matches essentially no StockCode; postage and similar rows therefore survive whenever they satisfy Quantity, UnitPrice and CustomerID. Scenario 2 also drops Country from the returned frame. Scenario 1 likewise does not keep Country. Hour in Scenario 2 is later read from Date (a calendar date, not a timestamp), so it collapses. These last two facts reappear in the causal chapter: the propensity model's categorical and cyclical covariates are degenerate.

3.4 Why both conventions must exist

If the 80,995-unit ticket is a system glitch, Scenario 1 double-counts a phantom and Scenario 2 is the truth. If it is a real order that was cancelled after the warehouse had already allocated stock, Scenario 1 is the exposure that risk management is supposed to see, and Scenario 2 is the accounting residual. A production WMS would resolve the ambiguity with an order-entry audit trail. On this public file one cannot. Reporting both scenarios is the substitute for that audit trail.

Quantity	Scenario 1 (gross)	Scenario 2 (netted)
Rows after cleaning	397,884	265,210
Unique SKUs	3,665	3,649
Gross / net merchandise value	£8,911,407.90	£8,307,997.38
Mean ticket value	£22.40	£31.33
Median ticket value	£11.80	£14.75
99th percentile value	£202.50	£316.15
99.9th percentile value	£876.00	£1,756.30
Maximum ticket	£168,469.60	£16,588.00
Mean quantity (unfiltered)	12.99	18.53
CV of quantity	13.807	5.010
Pearson kurtosis of quantity	178,189	3,324

Table 1. *The two working samples. Netting removes the £168k PAPER CRAFT ticket and collapses a large mass of cancellations; the remaining net tickets are fewer, larger on average, and still heavy-tailed.*

4. Global structure of demand: moments, Pareto, dependence, Gaussianity

4.1 Location, scale, shape

Let $Q_1,\ldots,Q_n$ be transaction quantities. The pipeline reports both the unfiltered sample and a Pareto-filtered subsample. Write

$$\bar{Q} = \frac{1}{n}\sum_{i=1}^{n} Q_i, \quad s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(Q_i - \bar{Q})^2, \quad \mathrm{CV} = \frac{s}{\bar{Q}}. \tag{4}$$

The coefficient of variation is dimensionless and is the natural scale-free dispersion for a positive variable. A CV of 13.8 (gross, unfiltered) means the standard deviation is an order of magnitude larger than the mean; the mean is not a typical ticket. Skewness and Pearson kurtosis are

$$\hat{\gamma}_1 = \frac{m_3}{s^3}, \qquad \hat{\kappa} = \frac{m_4}{s^4}, \qquad m_r = \frac{1}{n}\sum_{i=1}^{n}(Q_i - \bar{Q})^r. \tag{5}$$

pandas.Series.kurtosis returns *excess* kurtosis (Fisher); the code adds 3 to report Pearson kurtosis κ, which equals 3 for a Gaussian. On the unfiltered gross file κ = 178,189. That single number is the argument against every Gaussian risk measure that will be written down later.

4.2 The 80/20 filter

Let S be the set of SKUs and let $T_s = \Sigma_{i:\ \mathrm{SKU}(i)=s}\, Q_i$ be total quantity of SKU *s*. Rank SKUs so that $T_{(1)} \geq T_{(2)} \geq \ldots$ and form the cumulative share

$$C_k = \frac{\sum_{j=1}^{k} T_{(j)}}{\sum_{s \in S} T_s}. \tag{6}$$

The high-volume set is $\{s : C_{k(s)} \le 0.8\}$, i.e. the *largest* prefix of the ranked list whose cumulative share is at most 80% of total quantity (the code's *cum_freq <= 0.8*; it can undershoot 80% by one SKU). Tail estimators are then computed on the transaction-level quantities of those SKUs, not on the SKU totals. The filter is a concentration device: it asks whether the tail of *operationally important* SKUs is heavy, not whether a long tail of one-off novelty SKUs is heavy.

Implementation note — what the repository actually does.
Scenario 1's EVT filter ranks SKUs by *sum* of quantity. Scenario 2's EVT filter ranks SKUs by *maximum* single-order quantity and retains those accounting for 80% of the sum of those maxima (797 SKUs, n = 131,137 tickets). The two filters are not the same object. Scenario 2 is deliberately looking at SKUs capable of a large single net order, which is the relevant grain after netting. Table 2's Scenario 2 diagnostics are computed under the *sum* filter in *global_stats.py*; Table 3's Scenario 2 tail indices use the *max* filter in *run_evt_gpd*.

4.3 Serial dependence of daily totals

Let Y_t be total quantity on calendar day t, t = 1,…,T. The sample autocorrelation at lag k is

$$\hat{\rho}_k = \frac{\sum_{t=k+1}^{T} (Y_t - \bar{Y})(Y_{t-k} - \bar{Y})}{\sum_{t=1}^{T} (Y_t - \bar{Y})^2}. \tag{7}$$

Ljung–Box (1978) tests the joint hypothesis $\rho_1=\ldots=\rho_h=0$ with

$$Q_{\mathrm{LB}} = T(T+2)\sum_{k=1}^{h}\frac{\hat{\rho}_k^2}{T-k} \to \chi_h^2 \tag{8}$$

under white noise. With h = 10 the pipeline reports Q_{LB} = 136.48 (p ≈ 0) on gross daily totals and Q_{LB} = 40.76 (p ≈ 0) after netting. Daily volume is not white. EVT theorems are typically stated for i.i.d. observations or for stationary sequences with mixing; the practical consequence is that tail-index standard errors computed under i.i.d. are optimistic, and that a Hawkes chapter is not an ornament.

4.4 Three Gaussianity tests, and why they all reject

Kolmogorov–Smirnov. Standardise $Z_i = (Q_i - \bar{Q})/s$ and let F_n be the empirical distribution function of the Z_i. The KS statistic against the standard normal cdf Φ is

$$D_n = \sup_z |F_n(z) - \Phi(z)|. \tag{9}$$

The code calls scipy.stats.kstest on the z-scores. Because mean and variance are estimated, the tabulated KS p-value is the Lilliefors situation: the test is liberal (p-values too small). It does not matter here. D_n = 0.473

(gross) and 0.421 (netted); even a correctly sized test would reject at any conventional level.

Anderson–Darling. The AD statistic weights the tails of the empirical process,

$$A^2 = n\int_{-\infty}^{\infty} \frac{(F_n(x) - F_0(x))^2}{F_0(x)(1 - F_0(x))}\, dF_0(x). \tag{10}$$

statsmodels.stats.normal_ad returns $A^2 = \infty$ on this file. That is not a bug in the printout. With outliers on the order of 80,000 against a mean of 13, $F_0(x)(1-F_0(x))$ in the far tail is smaller than machine precision relative to the squared discrepancy, and the integral overflows. It is the correct qualitative message: the Gaussian tail is not a model for these data.

Shapiro–Wilk. The SW statistic is the squared correlation between ordered observations and normal scores, equivalently

$$W = \frac{\left(\sum_{i=1}^{n} a_i Q_{(i)}\right)^2}{\sum_{i=1}^{n} (Q_i - \bar{Q})^2}, \tag{11}$$

with $a = m^T V^{-1} / \|\cdot\|$ constructed from the mean and covariance of normal order statistics. W is computed on a without-replacement subsample of 5,000 (the Royston approximation is not reliable at $n \approx 4\cdot 10^5$). That subsample is drawn with the global NumPy RNG, not *default_rng(314159)*, so W is not seed-reproducible. On the recorded run W = 0.290 (gross) and 0.066 (netted); a Gaussian has $W \approx 1$.

4.5 A first look at the tail, before EVT

The empirical mean excess at the 99th percentile of the Pareto-filtered quantities is already a heavy-tail smoking gun. With $u = \hat{Q}_{0.99}$,

$$e_n(u) = \frac{\sum_{i=1}^{n} (Q_i - u)_+}{\#\{i : Q_i > u\}}. \tag{12}$$

One obtains $e_n(144) = 288.1$ (gross) and $e_n(276) = 469.8$ (netted). For an exponential random variable the mean excess is constant and equal to the scale; a mean excess twice the threshold is the signature of a regularly varying tail, which Section 5 formalises.

Diagnostic	Gross (S1)	Netted (S2)	Gaussian benchmark
Mean quantity (unfiltered)	12.99	18.53	—
CV	13.807	5.010	whatever σ/μ is
Skewness	409.89	43.69	0
Pearson kurtosis	178,189	3,324	3
ACF lags 1–5 (daily totals)	0.27, 0.15, 0.16, 0.16, 0.20	0.18, 0.15, 0.10, 0.14, 0.12	≈ 0
Ljung–Box Q (lag 10)	136.48 (p=0)	40.76 (p=0)	χ²_10
KS D (z-scores)	0.473 (p=0)	0.421 (p=0)	0
Shapiro–Wilk W (n=5,000)	0.290 (p=0)	0.066 (p=0)	≈ 1
99th pctile quantity	144	276	—
Mean excess at 99th pctile	288.1	469.8	≈ scale

Table 2. *Global diagnostics. Every Gaussian and every white-noise benchmark is rejected. The tail chapter is not optional. KS, Shapiro–Wilk, the 99th percentile and the mean excess use the Pareto-filtered quantity vector; mean, CV, skewness and kurtosis in the first four rows are unfiltered.*

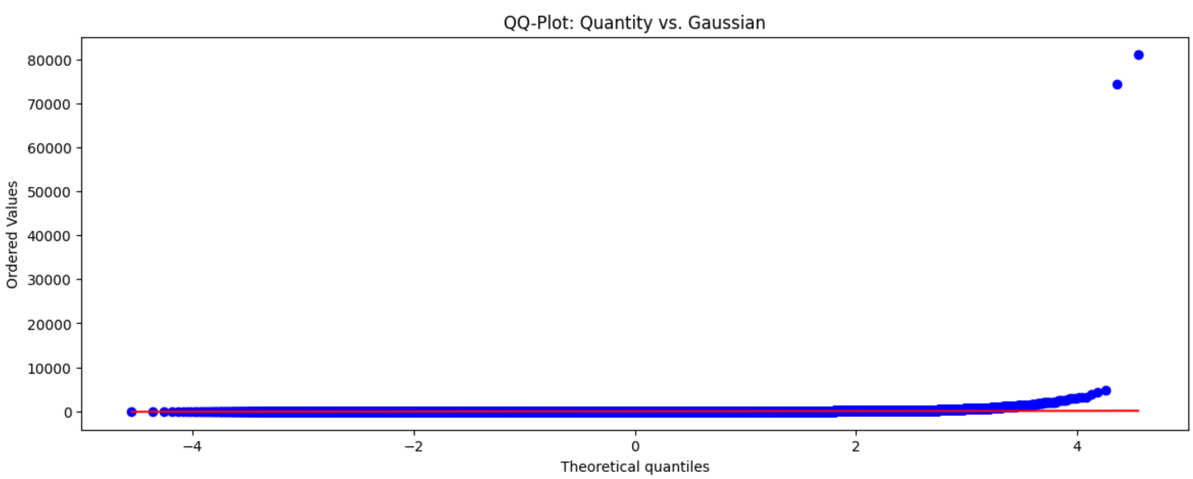

Figure 2. *Normal QQ-plot of Pareto-filtered quantities, Scenario 1. The right-tail bend is the Fréchet signature in a Gaussian plot.*

5. Extreme value theory, from first principles to the fitted tail

5.1 What EVT is for

Let $X_1,\ldots,X_n$ be i.i.d. with distribution function F, and write $M_n = \max\{X_1,\ldots,X_n\}$. For any F with a finite right endpoint or a thin tail, M_n is not interesting to a risk manager: sample maxima concentrate. For a heavy-tailed F, M_n grows and the way it grows is classified by a single real parameter ξ, the extreme-value index (tail index). EVT is the limit theory of M_n and, equivalently, of excesses $X - u$ given $X > u$ as u tends to the right endpoint. This section derives the three limit laws, the Hill estimator as the MLE of an exact Pareto tail, the Dekkers–Einmahl–de Haan moment estimator, GEV fitting of block maxima, the Pickands–Balkema–de Haan theorem, the GPD likelihood (used later on daily loss, not on quantity), the mean-excess diagnostic, an AMSE threshold diagnostic, and the empirical numbers on this file.

5.2 Regular variation

Definition (Regular variation at infinity).
A survival function $1 - F$ is regularly varying with index $-1/\xi$, $\xi > 0$, if there exists a slowly varying function L such that $1 - F(x) = x^{-1/\xi} L(x)$ for x large. Slowly varying means $L(tx)/L(x) \to 1$ as $x \to \infty$ for every $t > 0$. Equivalently, $(1 - F(tx))/(1 - F(x)) \to t^{-1/\xi}$. The number $\alpha = 1/\xi$ is the tail exponent: $P(X > x)$ behaves like $x^{-\alpha}$.

If $\xi > 1/2$ then $\mathrm{Var}(X) = \infty$; if $\xi \geq 1$ then $E[X] = \infty$. The Hill estimates on this file sit near $\xi \approx 0.5$, i.e. on the knife-edge of infinite variance. That is why a sample standard deviation of quantity is a fragile object, and why expected shortfall, which is a tail mean, must be handled through a GPD rather than through a sample mean of the whole distribution.

5.3 Fisher–Tippett–Gnedenko

Suppose there exist affine normalisations $a_n > 0$, b_n in *R* such that

$$\frac{M_n - b_n}{a_n} \;\to\; G \tag{13}$$

for a non-degenerate G. Then G belongs to one of three families, which can be written as a single generalised extreme value (GEV) law (von Mises–Jenkinson):

$$G_\xi(z) \;=\; \exp\left(-(1+\xi z)^{-1/\xi}\right), \qquad 1+\xi z > 0, \tag{14}$$

with the $\xi \to 0$ case defined by continuity as $G_0(z) = \exp(-e^{-z})$. The sign of ξ is the whole theory:

- **$\xi > 0$, Fréchet.** Heavy tails, regularly varying survival, unbounded support. Maxima grow like n^ξ. This is the empirically relevant case here.
- **$\xi = 0$, Gumbel.** Light tails of exponential type (Gaussian, log-normal, exponential, gamma). Mean excess of the parent is asymptotically constant or slowly varying.
- **$\xi < 0$, Weibull.** Finite right endpoint. Excesses shrink linearly toward that endpoint. A negative ξ on *daily loss* (Section 11, Scenario 2) is possible even when *quantity* is Fréchet, because daily loss is a sparse compound of a few unfulfilled dragons.

Theorem (Fisher–Tippett–Gnedenko, 1928/1943/1948).
The only possible non-degenerate limit laws of affinely normalised i.i.d. maxima are the GEV family $\{G_\xi : \xi \text{ in } R\}$. F is said to lie in the max-domain of attraction of G_ξ, written F in $\mathrm{MDA}(G_\xi)$.

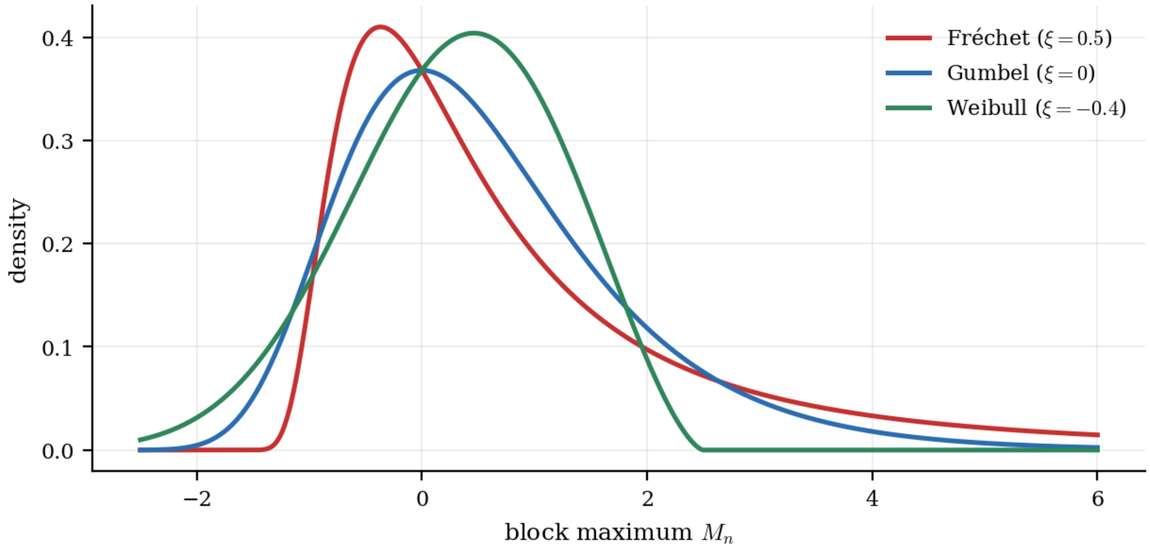

Figure 3. *GEV densities in the three domains, unit location and scale. Fréchet (red) has a polynomial right tail; Gumbel (blue) is exponential; Weibull (green) is bounded on the right.*

5.4 Why scipy's c is −ξ

scipy.stats.genextreme uses the convention of the National Institute of Standards, in which the shape is $c = -\xi$ and the cdf is $\exp(-(1 - c\,z)^{1/c})$ for $c \neq 0$. The pipeline therefore reports

$$\hat{\xi}_{\text{GEV}} \;=\; -\,\hat{c}_{\text{genextreme}}. \tag{15}$$

A positive printed ξ_GEV is a negative scipy c, i.e. Fréchet. This is the Coles (2001) sign convention, and it is the one used throughout this paper.

5.5 Hill's estimator, derived as a Pareto MLE

Condition on $X > u$ and suppose the *parent*, not the excess $Y = X - u$, is exactly Pareto with tail index ξ: $P(X > x \mid X > u) = (x/u)^{-1/\xi}$ for $x > u$. (The excess of a GPD is a different object, used later on daily loss.) The log-likelihood of the k observations $x_1,\ldots,x_k$ above u is

$$\ell(\xi) \;=\; -\,k\log\xi - k\log u - (1+\xi^{-1})\sum_{j=1}^{k}\log\frac{x_j}{u}. \tag{16}$$

The $-k \log u$ term does not depend on ξ. Differentiate the rest: $d\ell/d\xi = -k/\xi + \xi^{-2}\,\Sigma \log(x_j/u)$. Setting the score to zero yields the closed form

$$\hat{\xi}^{\text{Hill}}_k \;=\; \frac{1}{k}\sum_{j=1}^{k}\log\frac{X_{(n-j+1)}}{X_{(n-k)}}, \tag{17}$$

where $X_{(1)} \le \ldots \le X_{(n)}$ are the order statistics and $u = X_{(n-k)}$ is the (k+1)-st upper order statistic. This is Hill (1975). Under regular variation with index $-1/\xi$ and $k \to \infty$, $k/n \to 0$, one has $\xi^{\text{Hill}} \to_p \xi$, and $\sqrt{k}\,(\xi^{\text{Hill}} - \xi)$ is asymptotically normal with variance ξ^2 in the exact Pareto case.

The pipeline takes k = floor(sqrt(n)) with n the *transaction* count (n = 397,884 in Scenario 1, so k = 630), sorts the Pareto-filtered quantities in decreasing order, and evaluates

$$\hat{\xi}^{\text{Hill}} = \frac{1}{k}\sum_{i=1}^{k}\log\frac{X_{[i]}}{X_{[k+1]}}, \tag{18}$$

where X_{[i]} is the i-th decreasing value. Empirically ξ^{Hill} = 0.515 (gross) and 0.602 (netted). Both are strictly positive: Fréchet. The implied tail exponents are α = 1/ξ ≈ 1.94 (gross) and 1.66 (netted). In both cases α < 2, so a finite-variance model of quantity is not justified asymptotically.

5.6 The moment estimator of Dekkers, Einmahl and de Haan

Hill is designed for ξ > 0. The moment estimator (Dekkers, Einmahl and de Haan, 1989) is consistent for ξ in *R*. Write $M_n^{(1)}$ for the Hill sum and

$$M_n^{(2)} = \frac{1}{k}\sum_{j=1}^{k}\left(\log\frac{X_{(n-j+1)}}{X_{(n-k)}}\right)2. \tag{19}$$

For ξ > 0 the log-excesses are asymptotically exponential with mean ξ, so $M_n^{(1)} \to \xi$ and $M_n^{(2)} \to 2\xi^2$: the ratio $(M_n^{(1)})^2$ / $M_n^{(2)}$ converges in probability to 1/2, the second-moment correction vanishes, and the estimator is asymptotically Hill. For general ξ in *R* the estimator

$$\hat{\xi}^{\text{Mom}} = M_n^{(1)} + 1 - \frac{1}{2}\left(1 - \frac{(M_n^{(1)})^2}{M_n^{(2)}}\right)^{-1} \tag{20}$$

is the form implemented in the code (the factor ½(1 − ratio)$^{-1}$ is the standard second-moment correction). On this file ξ^{Mom} = 0.551 (gross) and 0.483 (netted). Agreement of Hill and moment, both positive, is the first robustness check that the Fréchet diagnosis is not an artefact of Hill's Pareto assumption.

5.7 Block maxima and a GEV fit

Partition the sample into m blocks of length b, b m ≈ n, and let $M_j^{(b)}$ be the maximum in block j. If F in MDA(G_ξ) then the block maxima are approximately GEV. The log-likelihood of an i.i.d. GEV sample is

$$\ell(\xi,\mu,\sigma) = \sum_{j=1}^{m}\left\{-\log\sigma - (1+\xi^{-1})\log\left(1+\xi\frac{M_j-\mu}{\sigma}\right) - \left(1+\xi\frac{M_j-\mu}{\sigma}\right)^{-1/\xi}\right\} \tag{21}$$

for ξ ≠ 0, with the Gumbel limit obtained by L'Hôpital. Maximisation is numerical (scipy.stats.genextreme.fit uses a combination of method of moments and MLE).

Implementation note — what the repository actually does.

The code sets block_size = 30 and walks through the *quantity array of high-volume SKUs in file order*, not through calendar months. The comment says "monthly blocks"; the implementation is blocks of 30 consecutive filtered transactions. In Scenario 1 the loop bound is n = len(df) rather than len(quantities), so empty slices at the end are skipped by a length check. The fitted ξ_GEV = 0.473 (gross) and 0.833 (netted). The netted GEV is the most dispersed of the three estimators, which is expected: block maxima of size 30 on a heavy tail are noisy, and Scenario 2's filter on maxima inflates block extremes.

5.8 Peaks over threshold: Pickands–Balkema–de Haan

Block maxima throw away all but one observation per block. The peaks-over-threshold (POT) method keeps every excess over a high threshold u. Let $x_F = \sup\{x : F(x) < 1\}$ be the right endpoint (possibly $+\infty$).

> **Theorem (Pickands–Balkema–de Haan).**
> F in MDA(G_ξ) if and only if there exists a positive measurable function σ(u) such that the conditional excess distribution $F_u(y) := P(X - u \le y \mid X > u)$ satisfies $\sup_{y \text{ in } (0,\, x\_F-u)} | F_u(y) - G_{\xi,\sigma(u)}(y) |$ -> 0 as u -> x_F, where $G_{\xi,\sigma}$ is the generalised Pareto distribution (GPD).

The GPD cdf, density and survival, for $y \ge 0$ and $1 + \xi\, y/\sigma > 0$, are

$$G_{\xi,\sigma}(y) = 1 - (1 + \xi y/\sigma)^{-1/\xi}, \quad g_{\xi,\sigma}(y) = \sigma^{-1}(1 + \xi y/\sigma)^{-1/\xi - 1}. \tag{22}$$

For ξ -> 0 this is the exponential law with mean σ. For ξ > 0 it is Pareto type II (Lomax) with unbounded support. For ξ < 0 it has a finite endpoint y = −σ/ξ. The scale used in the code for daily-loss POT is written β rather than σ; this paper uses β_u for the GPD scale to avoid clashing with Hawkes decay.

5.9 Mean excess, and why a positive slope is the GPD's signature

The mean excess function e(u) = E[X − u | X > u], when it exists (ξ < 1), is, for an exact GPD,

$$e(u) = \frac{\beta_u}{1-\xi}, \qquad \beta_u = \beta + \xi(u - u_0) \qquad (\xi < 1). \tag{23}$$

Here β_u is the GPD scale of excesses over u, as in the code, and β is that scale at a reference threshold u_0. Substituting the linear scale gives the affine form $e(u) = (\beta + \xi(u-u_0))/(1-\xi)$, with slope ξ/(1−ξ). Writing $(\beta_u + \xi u)/(1-\xi)$ would double-count the ξu term. A scatter of empirical mean excesses against u that is linearly increasing in the upper tail is therefore visual evidence for ξ > 0. The pipeline computes

$$e_n(u) = \frac{\sum_i (X_i - u)_+}{\#\{i : X_i > u\}} \tag{24}$$

on a grid of empirical percentiles from the 90th to the 99th (40 points), then fits a least-squares slope on the upper 40% of that grid. The fitted slopes are +1.782 (gross) and +1.089 (netted). Both are positive. If one treats the slope as ξ/(1−ξ), one recovers a rough ξ ≈ 0.64 (gross) and 0.52 (netted), in the same neighbourhood as Hill. This is a diagnostic, not an estimator: the least-squares slope on a dependent grid is not Hill.

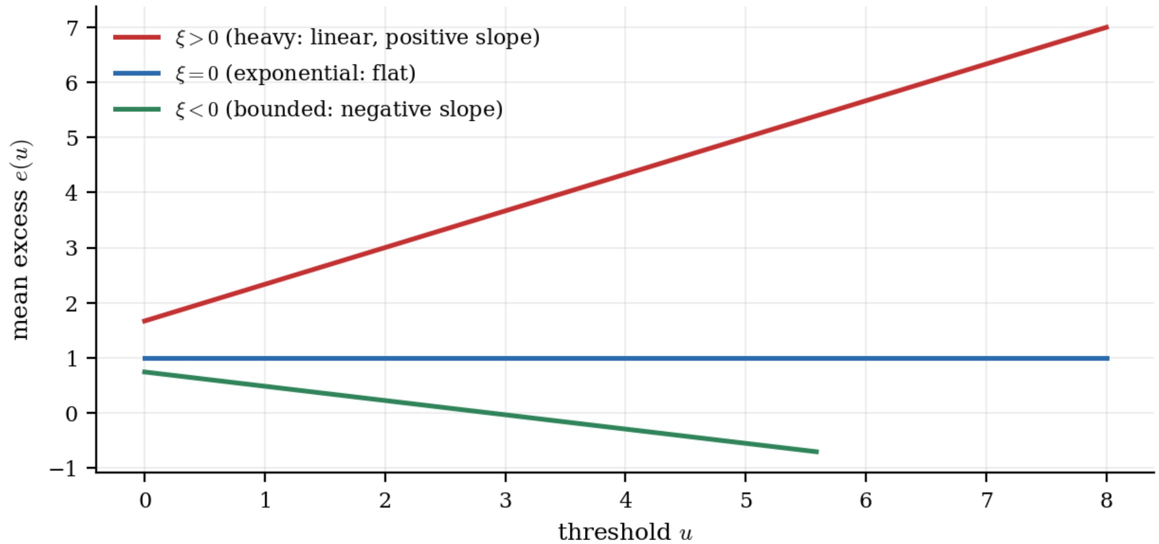

Figure 4. *Theoretical mean-excess functions of a GPD with unit scale. Only the Fréchet case (ξ > 0) produces a linearly increasing mean excess. That is the pattern in both empirical plots of the repository.*

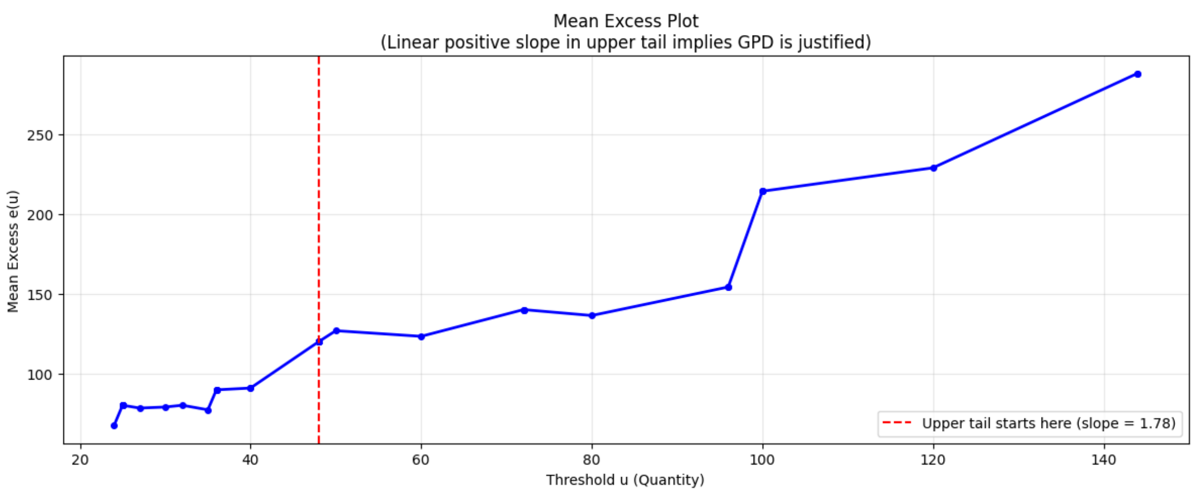

Figure 5. *Empirical mean-excess plot, Scenario 1, Pareto-filtered quantity. The upper-tail least-squares slope is +1.782.*

5.10 GPD maximum likelihood

Given exceedances $Y_j = X_j - u > 0$, $j = 1,\dots,N_u$, the GPD log-likelihood is

$$\ell(\xi, \beta_u) = -N_u \log \beta_u - \left(1 + \tfrac{1}{\xi}\right) \sum_{j=1}^{N_u} \log\left(1 + \xi \frac{Y_j}{\beta_u}\right), \tag{25}$$

provided $\beta_u > 0$, $\xi > -1$ (so the GPD density is defined on the excesses), and $1 + \xi\, Y_j/\beta_u > 0$ for all j. Asymptotic normality of the MLE additionally wants $\xi > -1/2$. Expected shortfall of the parent is finite only for $\xi < 1$. The negative log-likelihood implemented for daily losses in the backtest module is exactly $-l$, with a barrier that returns a large constant when $\beta_u \le 0$, $\xi \ge 1$, or any term $1 + \xi\, Y/\beta_u$ is non-positive, and with box bounds ξ in [−0.5, 1]. The code *rejects* $\xi \ge 1$: ES of a GPD parent is infinite there, so the optimiser is kept out of the non-integrable region. This likelihood is *not* fitted to quantity. With $N_u = 4$ (Scenario 1 daily losses) this MLE is a four-point fit of a two-parameter model; Section 11 returns to that fragility.

5.11 AMSE-optimal threshold

Threshold choice is the practical problem of POT. Too low a u contaminates the GPD approximation (bias); too high a u starves the likelihood (variance). The code does *not* recompute the Hill estimator (17) on original values above u. For each candidate threshold it forms the excesses $Y = X - u$, sets $k = N_u$ (every excess), and evaluates a Hill-type mean of $\log(Y_{[i]} / Y_{min})$ on i.i.d. bootstrap resamples of those excesses. A bootstrap estimate of asymptotic mean squared error is

$$\widehat{\mathrm{AMSE}}(u) \;=\; \left(\bar{\xi}^*_u \;-\; \hat{\xi}^{\mathrm{Hill}}\right)^2 \;+\; \widehat{\mathrm{Var}}(\xi^*_u), \tag{26}$$

where ξ*_u is this *excess-Hill* on a bootstrap resample of $\{X - u : X > u\}$, the bar is a bootstrap mean (B = 200), and the Hill on the right is still the full-sample Hill of Section 5.5, used as a bias anchor. Because the denominator is the smallest excess, the statistic is numerically fragile when a bootstrap draw includes a near-zero Y. The code evaluates this on the percentile grid {92, …, 98.5} (25 points), skips any u with fewer than 50 excesses, and takes u* = arg min AMSE(u). Empirically u* = 96.0 (gross, which happens to equal the 97.5th percentile of filtered quantity) and u* = 49.0 (netted).

Three caveats. First, the bias is measured against full-sample Hill, not against a truth; AMSE here is a stability criterion, not an oracle MSE. Second, the bootstrap of excesses is i.i.d. and ignores the daily dependence documented in Section 4.3. Third, and operationally decisive: *u* is never passed to a GPD MLE on quantity*. The quantity chapter is a Fréchet-domain diagnosis. The only GPD likelihood in the repository is the daily-loss fit of Section 11. None of this overturns the qualitative conclusion that a regularly varying tail, not a Gaussian, is the right family for quantity.

Implementation note — what the repository actually does.
Scenario 1 indexes *opt_u = u_grid_amse[argmin(amse_vals)]* on the full percentile grid after *continue* on $k_u < 50$, so a skipped u would misalign the minimiser. Scenario 2 keeps a parallel *valid_u* list and is safe. On the recorded run no skip fired in Scenario 1 (u* sits on the 97.5th percentile of filtered quantity).

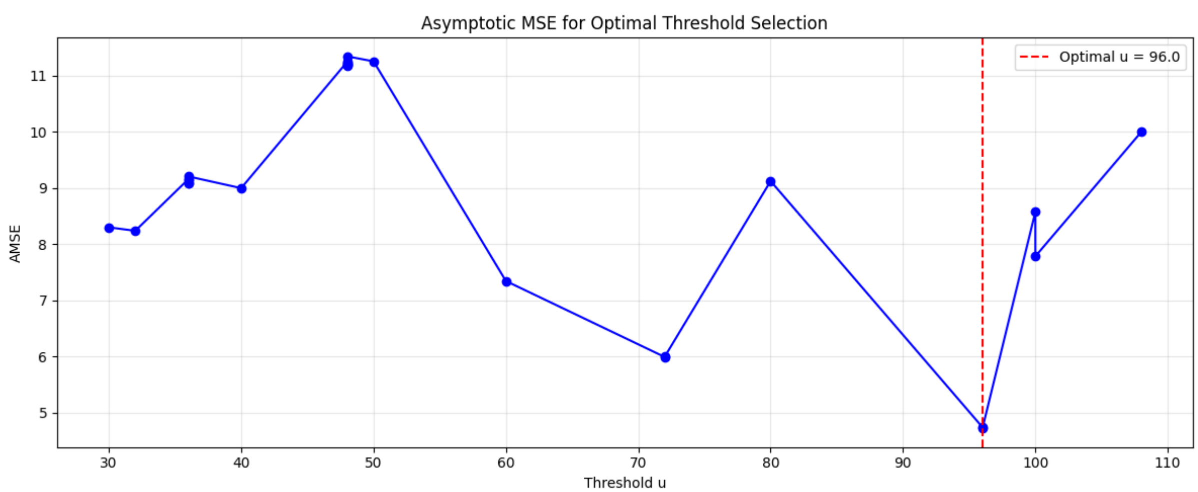

Figure 6. *Bootstrap AMSE of the excess-Hill statistic as a function of threshold, Scenario 1. The vertical line is the minimiser u* = 96. That threshold is not used to fit a GPD on quantity.*

5.12 Empirical tail-index table, and the Fréchet conclusion

Estimator	S1 gross	S2 netted	What it is
Hill ξ (k = floor(sqrt(n)))	0.515	0.602	MLE of an exact Pareto tail
Moment ξ	0.551	0.483	Dekkers–Einmahl–de Haan
GEV block-max ξ	0.473	0.833	MLE on 30-transaction blocks
Mean-excess slope	+1.782	+1.089	LS on upper 40% of $e_n(u)$
Implied ξ from slope	≈ 0.64	≈ 0.52	slope = ξ/(1−ξ)
AMSE threshold u*	96.0	49.0	min excess-Hill AMSE (unused downstream)
Domain	Fréchet	Fréchet	all ξ > 0 on quantity

Table 3. *Tail-index estimates on high-volume SKU quantities. Quantity is in the Fréchet domain in both scenarios. Daily-loss GPD in Section 11 is a different random variable and can have a different ξ.*

The paper's EVT conclusion is therefore modest and firm. Transaction quantity, on the SKUs that constitute 80% of volume (S1) or 80% of max-order mass (S2), has a regularly varying right tail with ξ in a neighbourhood of one half. Gaussian, exponential, and bounded-tail models for quantity are not compatible with this evidence. That is the licence for everything that follows: a dragon is not an outlier to be winsorised; it is a point in a regularly varying tail, and the loss it can generate has to be read with tail tools.

Remark.
Convergence of three estimators is not a proof of i.i.d. regular variation. It is a check that the Fréchet diagnosis is not an artefact of one functional. Dependence (Section 4.3), the 30-transaction GEV blocking, and the value-biased sampling of dragons later in the pipeline all mean that ξ should be read as a directional tail index, not as a number to three decimals for a regulator.

6. Synthetic delays, dragon assignment, holding cost, and SLA loss

6.1 Why the delays are synthetic

InvoiceDate is a billing timestamp, not a fulfilment clock. There is no pick-start, no pack-complete, no carrier handover, and no SLA flag in the UCI file. Any delay attached to a ticket is therefore a random variable constructed by the analyst. The construction is parametric and documented in *src/config.py*. Parameters are educated values informed by public summaries of the 2025 WERC DC Measures Report and the CSCMP State of Logistics Report, not extractions from those proprietary files. This section writes the construction down so that the invented randomness is fully specified.

6.2 Value-biased dragon assignment

Let n be the number of tickets and let p_d be the dragon rate (0.00032 gross, 0.00025 netted). Let γ be the bias exponent (1.35 gross, 1.20 netted). Define sampling weights

$$w_i = \frac{V_i^\gamma}{\sum_{j=1}^{n} V_j^\gamma}. \tag{27}$$

A set D of size n_d = floor(p_d n) is drawn without replacement from {1,…,n} with probabilities w. (numpy's choice with p=w and replace=False implements a sequential weighted sample; inclusion probabilities are not exactly w_i, but they are monotone in V_i and approximately w_i for small p_d.) The remaining tickets contribute a surge set S of size floor(0.18 n) drawn uniformly, and the rest are labelled normal. Write T_i in {normal, surge, dragon} for the resulting tier. Empirically |D| = 127 (0.0319%) in Scenario 1 and 66 (0.0249%) in Scenario 2.

Proposition (Stochastic dominance, mechanical).
If $\gamma > 0$ and sampling is value-biased as above, then V | T=dragon likelihood-ratio dominates V | T≠dragon in the large-n weighted-sampling limit. Consequently Q_q(V | dragon) − Q_q(V | non-dragon) ≥ 0 at every quantile q, with strict inequality on a set of q of positive measure. The QTE of Section 8 is therefore partly a restatement of the assignment rule. It is still the right descriptive contrast for a risk manager, because the assignment rule was chosen to put the expensive tail in the dragon bucket; it is not an estimated effect of an external treatment.

6.3 Log-normal delays, parameterised by the mean

Given a tier, delay D (minutes) is drawn independently from a log-normal and then clipped to a closed interval. A log-normal is specified by (μ_l, σ_l), but the configuration file specifies a *mean in minutes* m and a log-scale σ_l. The translation is the elementary moment identity. If log $D^0 \sim N(\mu_l, \sigma_l^2)$ then

$$\mathbb{E}[D^0] = \exp\left(\mu_\ell + \tfrac{1}{2}\sigma_\ell^2\right), \qquad \mathrm{Var}(D^0) = \left(e^{\sigma_\ell^2} - 1\right)\left(\mathbb{E}[D^0]\right)^2. \tag{28}$$

Imposing $E[D^0]$ = m gives $\mu_l = \log m - ½\,\sigma_l^2$, which is exactly the line in *delays.py*:

$$\mu_\ell = \log m - \tfrac{1}{2}\sigma_\ell^2, \qquad D = \min(b, \max(a, \exp(\mu_\ell + \sigma_\ell Z))), \tag{29}$$

with Z ~ N(0,1). Clipping is not innocent: it truncates the log-normal and therefore lowers the realised mean relative to m, more so for dragons (σ_l = 0.85, clip [180, 1440] minutes). The configured means are targets of the unclipped variable, not of the clipped one.

Tier	Share	Target mean m	σ_l	Clip (min)	S1 realised mean
normal	≈ 82%	25 min	0.50	[5, 120]	25.0 min
surge	18%	90 min	0.65	S1 [30, 480]; S2 [30, 360]	90.1 / 90.0 min
dragon	32 bps / 25 bps	420 / 360 min	0.85 / 0.82	S1 [180, 1440]; S2 [200, 1440]	389.9 / 302.8 min

Table 4. *Delay configuration (config.py) against realised means. Dragon realised means sit below the unclipped target because of the right clip at 24 hours and the left clip at 3–3.3 hours.*

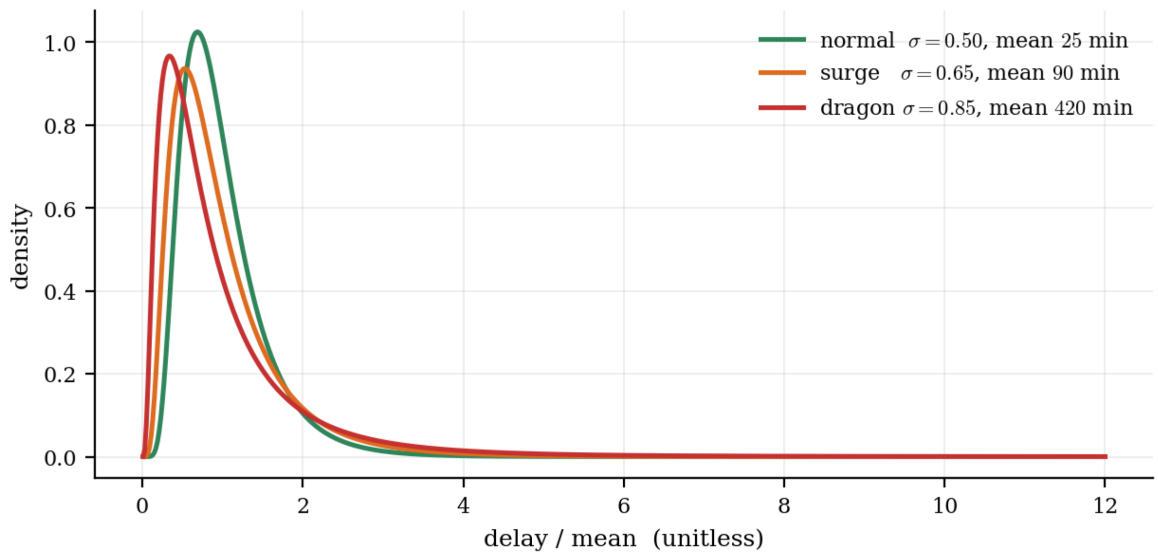

Figure 7. *Shape of the unclipped log-normal for the three configured σ values, plotted in units of the mean. Larger σ is a heavier relative tail; clipping (not shown) then cuts both ends.*

6.4 SLA bins and the unfulfilled-dragon indicator

SLA status is a deterministic function of delay, via pd.cut(..., right=False) on the bins [0, 30, 120, 480, ∞) minutes in Scenario 1 and [0, 30, 120, 360, ∞) in Scenario 2, labelled On-Time, Minor Delay, SLA Breach, Fulfilment Failure. Because the cuts are left-closed, a delay of exactly 30 minutes is Minor Delay, not On-Time. An unfulfilled dragon is a dragon whose delay also clears an explicit breach threshold τ (240 min = 4 h in S1; 360 min = 6 h in S2) and whose SLA label is one of the two breach labels:

$$U_i = \mathrm{I}(T_i = \text{dragon})\, \mathrm{I}(D_i \geq \tau)\, \mathrm{I}(\mathrm{SLA}_i \in \text{breach or failure}). \tag{30}$$

On the recorded run this identifies 66 of 127 dragons (52.0%) in Scenario 1 and 17 of 66 (25.8%) in Scenario 2. The Scenario 2 rate is lower both because τ is higher (6 h vs 4 h) and because the dragon mean delay is lower (360 vs 420). Those two modelling choices are the reason Scenario 2 is the “realistic” view in the README, not any property of netting itself—although netting also removes the £168k ticket, which would otherwise dominate unfulfilled revenue.

6.5 Holding cost

Working-capital drag is charged at an annual holding rate h = 0.25. Delay in days is $D_i/1440$, so

$$H_i = V_i \cdot \frac{D_i}{1440} \cdot h, \qquad R_i^{\text{net}} = V_i - H_i. \tag{31}$$

This is a linear carrying-cost identity, not an inventory model: there is no (Q, r) policy, no cycle stock, no obsolescence. Summing over the file gives holding of £75,635.58 (0.849% of gross) in Scenario 1 and £59,274.45 (0.713%) in Scenario 2. Holding is a small drag. The loss that matters is the unfulfilled-dragon revenue $\Sigma_i U_i V_i$, equal to £147,582.95 (S1) and £31,158.62 (S2).

Remark.
Two different "revenue at risk" numbers are printed in the delay stage. One is the sum of V_i over all tickets whose SLA is breach or failure, including non-dragons (£694,291 in S1, 7.791% of gross). The other is the unfulfilled-dragon sum (£147,583). The Monte Carlo, causal annual impact, and Hawkes P&L all use the *dragon* sum. The broader SLA sum is a dashboard statistic, not the loss functional of Sections 7–10. Section 13's reconciliation table keeps them on separate rows.

7. Monte Carlo value-at-risk and expected shortfall

7.1 VaR and ES, as functionals

Definition (Value-at-risk and expected shortfall).
Let L be a loss random variable with distribution function F_L. For α in (0,1) (here α = 0.95 or 0.99), $\mathrm{VaR}_\alpha(L) = \inf\{l : F_L(l) \geq \alpha\} = F_L^{<-}(\alpha)$. Expected shortfall is $\mathrm{ES}_\alpha(L) = (1/(1-\alpha)) \int_\alpha^1 \mathrm{VaR}_u(L)\,du$. If F_L is continuous, this equals $E[L \mid L \geq \mathrm{VaR}_\alpha(L)]$. ES is coherent (Artzner et al., 1999); VaR is not, because it can penalise diversification. The pipeline's headline is $\mathrm{ES}_{0.95}$.

On a Monte Carlo sample $L^{(1)},\ldots,L^{(P)}$ of size P = 10,000, the estimators are the empirical quantile and the tail mean:

$$\widehat{\mathrm{VaR}}_\alpha = L_{(\lceil \alpha P\rceil)}, \qquad \widehat{\mathrm{ES}}_\alpha = \mathrm{mean}(L^{(p)} \mid L^{(p)} \geq \widehat{\mathrm{VaR}}_\alpha). \tag{32}$$

That is the order-statistic convention. The code does not take $L_{(\lceil \alpha P\rceil)}$. It calls *numpy.percentile*, which linearly interpolates (Hyndman–Fan type 7). For P = 10,000 the two conventions differ by a fraction of a path; they are not identical. The expected-shortfall estimator is the sample mean of paths at or above that percentile, matching the code.

7.2 The simulated annual loss

Let V_d be average daily dragon *revenue*, computed as total dragon revenue divided by the number of calendar days in the sample (including days with no dragon):

$$\bar{V}_d = \frac{1}{\Delta}\sum_{i\in\mathcal{D}} V_i, \qquad \Delta = 374. \tag{33}$$

Let r_V be the revenue breach rate, the share of dragon revenue that sits on unfulfilled dragons, and let g = 0.30 be the gross-margin factor:

$$r_V = \frac{\sum_i U_i V_i}{\sum_{i\in\mathcal{D}} V_i}, \qquad g = 0.30. \tag{34}$$

Daily dragon revenue on a simulated day is taken log-normal with mean V_d and coefficient of variation equal to the CV of *dragon-active days only*, capped at 3:

$$\mathrm{CV} \;=\; \min\Big\{3,\; \frac{s_{\text{active}}}{\bar V_{\text{active}}}\Big\}. \tag{35}$$

Method of moments for a log-normal R with E[R] = V_d and CV as above:

$$\sigma_d^2 = \log(1+\mathrm{CV}^2), \qquad \mu_d = \log \bar V_d - \tfrac{1}{2}\sigma_d^2, \qquad R_t = \exp(\mu_d + \sigma_d Z_t), \quad Z_t \sim_{\text{iid}} N(0,1). \tag{36}$$

A path p = 1,…,P is a year of such days, converted to margin-adjusted preventable loss by a constant factor:

$$L^{(p)} \;=\; g\,r_V \sum_{t=1}^{365} R_t^{(p)}. \tag{37}$$

7.3 The mean of L is not the printed EL

Take expectations in the path equation. E[R_t] = V_d = (Σ_D V_i)/Δ, so

$$\mathbb{E}[L] = g\,r_V \cdot 365 \cdot \bar V_d = g \cdot \frac{365}{\Delta} \sum_i U_i V_i. \tag{38}$$

The quantity the Monte Carlo module prints as "Expected Loss (EL) — data-derived" is the *gross* annualisation (365/Δ) Σ U_i V_i, without g. Call it EL_{gross}. Then E[L] = g · EL_{gross}. On Scenario 1, EL_{gross} = £144,031 and g · EL_{gross} = £43,209, which is exactly the causal module's "total annual impact" and sits next to the Monte Carlo median of £42,387 (right-skew: mean above median). The headline $ES_{0.95}$ = £60,134 is a tail functional of L, i.e. already margin-adjusted. Comparing £144,031 to £60,134 as if they were estimates of the same thing is a category error; Section 13 tabulates this.

7.4 Unexpected loss

The printed "unexpected loss" is the gap between $VaR_{0.95}$ and the median of L, UL = $VaR_{0.95}$ − median(L), equal to £12,192 (S1) and £1,968 (S2). This is a dispersion measure of the simulated annual P&L, not a regulatory UL from a credit-risk IRB formula. It is small relative to EL_{gross} because the i.i.d. log-normal year-sum, with CV capped at 3, is not a disaster distribution once it has been averaged over 365 days. Clustering (Section 9) would fatten this UL if it were folded into the paths; it is not.

Functional of L (margin-adjusted unless noted)	Scenario 1	Scenario 2
$EL_{gross} = (365/\Delta)\ \Sigma\ U_i V_i$ (not a mean of L)	£144,031	£30,409
$g \cdot EL_{gross} = E[L]$ (approx.)	£43,209	£9,123
Median of L	£42,387	£9,021
$VaR_{0.95}(L)$	£54,579	£10,989
$VaR_{0.99}(L)$	£62,998	£12,062
$ES_{0.95}(L)$ (MC module headline)	£60,134	£11,716
$UL = VaR_{0.95} - \text{median}$	£12,192	£1,968
Active-day CV (capped at 3)	3.00	2.28
Paths × horizon	10,000 × 365	10,000 × 365
Seed	314159	314159

Table 5. *Monte Carlo module outputs. ES95 here is the headline of src/risk/monte_carlo.py. It is not the last-window backtest ES95 of Section 12.*

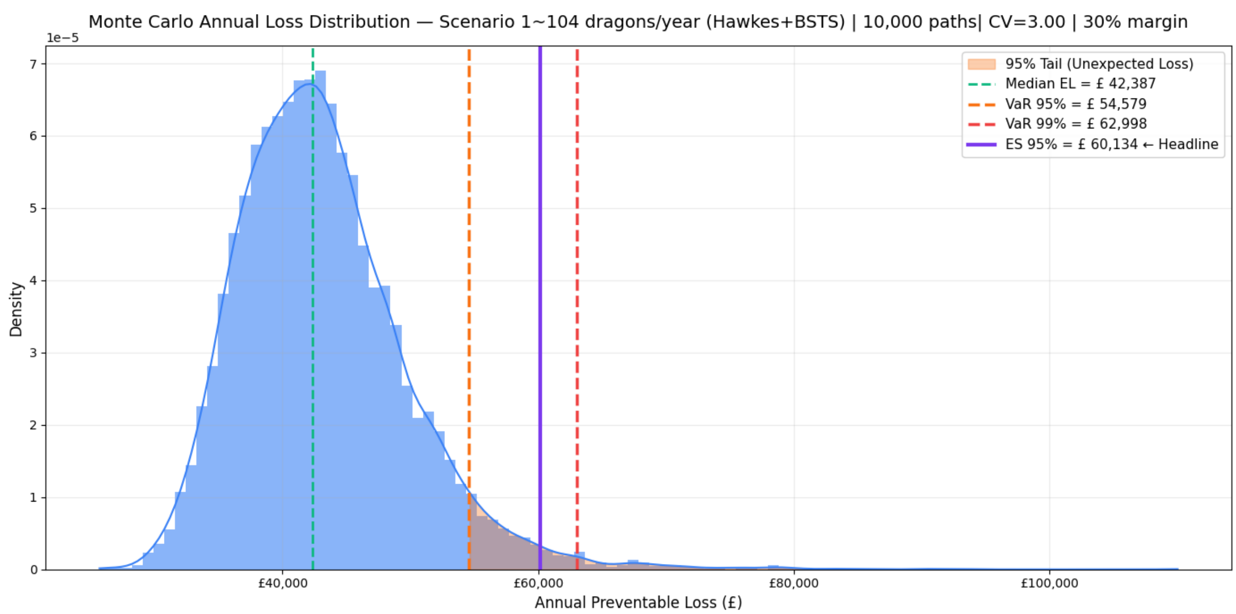

Figure 8. *Monte Carlo annual loss density, Scenario 1, P = 10,000. Vertical lines are median, VaR95, VaR99 and ES95 of L (margin-adjusted). The embedded plot title still says "Hawkes+BSTS"; that is a dashboard leftover. The paths are i.i.d. log-normal, and the count projection in Section 10 is a Kalman filter, not a Bayesian BSTS.*

Implementation note — what the repository actually does.

Daily revenues are i.i.d. log-normal. Hawkes clustering is estimated on the same dragons and never injected into these paths (no thinning, no stochastic intensity). The CV is computed on dragon-active days but the mean V_d is computed over all calendar days, so the log-normal is a mixture-mean with an active-day dispersion: a slightly incoherent pair, conservative on dispersion if many days are zeros. The backtest module (Section 12) uses a different mean—the mean of active days only—which is why its ES95 is several times larger. TARGET_ANNUAL_DRAGONS in config.py (104 / 72) is printed in this module and is not used in the path equation.

8. Causal contrasts: ATE as negative control, QTE as the object of interest

8.1 What is being identified

There are two binary flags, and they must not be conflated. The *surge* flag $W_i = 1\{i \text{ in } S\}$ is assigned uniformly among non-dragons. The *dragon* flag $\Delta_i = 1\{i \text{ in } D\}$ is assigned with weights V_i^{γ}. The causal engine's propensity-score and quantile-regression machinery treats W as a treatment and net revenue R^{net} as an outcome. The dragon machinery treats Δ as a grouping variable and V as an outcome. Only the second grouping is economically interesting, and it is not a treatment in the Neyman–Rubin sense.

8.2 Why the surge ATE must be near zero

Potential outcomes $(R^{net}(1), R^{net}(0))$ for surge would be well-defined if surge were an intervention that changed delay and, through delay, holding cost. In the simulator that is true in Scenario 1: W_i changes the delay mean from 25 min to 90 min, so H_i rises by $V_i \cdot (65/1440) \cdot 0.25 \approx 0.011\, V_i$, about 110 basis points of value. The average treatment effect on net revenue is therefore on the order of minus one percent of average ticket value, i.e. a few tens of pence, plus noise from matching. There is no path from W to V, because V is determined before assignment. Empirically the caliper-matched ATE is −£2 (S1) and −£18 (S2), and quantile regression coefficients on W are £0 at almost every quantile, with a −£7 coefficient at the 99.9th in Scenario 1. That is a passed negative control, not a finding about warehouse physics. (Scenario 2's PSM/QR outcome is order value, not V − H; see the implementation note in 8.3.)

8.3 Propensity scores, in theory

Rosenbaum and Rubin (1983): if unconfoundedness Y(0),Y(1) independent of W given X and overlap $0 < P(W=1|X) < 1$ hold, then Y(0),Y(1) independent of W given π(X) where $\pi(x) = P(W=1 \mid X=x)$. A logit model

$$\pi(x) = \frac{\exp(x^\top \theta)}{1 + \exp(x^\top \theta)} \tag{39}$$

is fitted by L2-penalised maximum likelihood (logistic regression, saga solver, C=1) on a without-replacement subsample of 50,000 tickets. Matching with caliper c = 0.005 keeps a treated unit t and a control u only if $|\pi_t - \pi_u| \le c$, within country. Scenario 2 is one-to-one without replacement on controls; Scenario 1 is not (see the implementation note). The matching ATE is

$$\widehat{\text{ATE}}_{\text{PSM}} = \frac{1}{|\mathcal{M}|} \sum_{(t,u) \in \mathcal{M}} \left(R_t^{\text{net}} - R_u^{\text{net}}\right). \tag{40}$$

Implementation note — what the repository actually does.

The loader does not return Country. The causal engine therefore fills Country = 'Unknown' for every row. Matching "within country" is matching within a single stratum. In Scenario 2, Hour is parsed from a calendar date (midnight), so it is identically 0, not a genuine hour-of-day. The propensity covariates are effectively (log quantity, a constant). The propensity range in Scenario 2 is [0.1975, 0.2072]—almost degenerate—which is exactly what a weak covariate set produces. The logit is not fit on all n tickets. After building a treated-plus-four-to-one control pool, the code draws a without-replacement slice of size PSM_SUBSAMPLE and maximises the penalised likelihood on that slice; scores are then predicted for every row. Scenario 1 matching is a within-bin Cartesian join of the full treated and control sets, then *drop_duplicates* on the treated propensity: the same control can match several treated units. Scenario 2 walks sorted propensity scores with a *c_used* flag and is genuinely one-to-one without replacement. The recorded run uses 50,000 (the log line is "Propensity fitted (n=50,000)"). A 250,000-row fit does not complete on this file: saga IRLS on that slice, followed by the full-sample caliper merge, exhausts time and memory and returns no ATE. Enlarging the fit sample would not change the match in any case, because matching is already full-sample. Fifty thousand is the size at which Table 6's surge ATE is actually produced. Scenario 2 also sets *Net_Revenue_GBP = OrderValue_GBP* and never subtracts holding. The "110 bp holding-cost ATE" calculation of Section 8.2 therefore applies only to Scenario 1. In Scenario 2 the PSM/QR outcome is order value, which is determined before assignment, so the true surge ATE is 0 and the recorded −£18 is matching noise. No standardised-mean-difference table, no love plot, no double-robust AIPW. LIMITATIONS.md already says this. For a negative control, degeneracy and the smaller fit sample are not fatal: we are checking that a near-null stays near-null.

8.4 Quantile regression

Koenker and Bassett (1978). For q in (0,1) the check function $\rho_q(u) = u\,(q - 1\{u < 0\})$ yields

$$\hat{\beta}(q) \;=\; \arg\min_{\beta} \sum_{i=1}^{n} \rho_q\left(Y_i - X_i^{\top}\beta\right). \tag{41}$$

The coefficient on W at quantile q is a q-quantile treatment effect under a linear model for the conditional quantile of net revenue. Fitted at q in {0.5, 0.75, 0.95, 0.99, 0.999} on a subsample of 180,000 (config currently lists [0.95, 0.999]; the recorded run used the longer list). All estimated coefficients are £0 to the printed precision, except q = 0.999 in Scenario 1 (−£7, $p = 2.1 \times 10^{-23}$). The p-value at the far quantile should not be over-read: the check-function asymptotics are delicate in the far tail, the subsample is not clustered by day, and the true ATE is a holding-cost effect of pennies to pounds.

8.5 Quantile treatment effects of the dragon label, the object of interest

Definition (QTE of dragon on value).

The q-quantile treatment effect of the dragon grouping on order value is $\delta(q) = F^{-1}_{V|\Delta=1}(q) - F^{-1}_{V|\Delta=0}(q)$. It is estimated by the difference of sample quantiles. It is a descriptive contrast, not an ATE of a manipulable treatment, because Δ is a function of V.

The dragon premium is the mean analogue, $\pi = E[V \mid \Delta=1] - E[V \mid \Delta=0]$. On the recorded run:

Contrast	Scenario 1	Scenario 2
Dragon mean premium π	£2,837	£1,700
QTE 50%	£167	£290
QTE 75%	£714	£1,784
QTE 95%	£3,675	£7,665
QTE 99%	£67,046	£13,409
QTE 99.9%	£156,138	£12,069
PSM ATE of surge on net revenue	−£2	−£18
QR coefficient on surge, q=0.5 … 0.99	£0	£0

Table 6. *Causal-engine outputs. The surge ATE/QR block is the negative control. The dragon premium and QTE block is the object of interest, and is partly mechanical from value-biased sampling. Scenario 2's 99.9th QTE sitting below the 99th is a small-sample quantile crossing (66 dragons). Table 6 uses the recorded-run quantile list {0.5, 0.75, 0.95, 0.99, 0.999}; current config.py keeps only [0.95, 0.999] and will not reprint the 50/75/99% rows.*

The QTE profile in Scenario 1 is the picture the project is built to show. At the median the dragon ticket is only a few hundred pounds richer than a non-dragon; at the 95th it is £3.7k richer; at the 99th, £67k; at the 99.9th, £156k. That is a regularly varying value tail, flagged by construction, and it is why a 4-hour SLA miss on a dragon is an entirely different economic event from a 4-hour miss on a median ticket. Scenario 2's QTE profile is flatter in the far tail because netting has removed the £168k PAPER CRAFT observation; the 99.9th crossing below the 99th is what 66 points do to an extreme quantile, and is reported honestly.

8.6 Annual impact, as used in the causal dashboard

The causal module's "total annual impact" is not an ATE. It is the margin-adjusted, calendar-annualised unfulfilled-dragon revenue,

$$I = g \cdot \frac{365}{\Delta} \sum_i U_i V_i = g \cdot \mathrm{EL}_{\mathrm{gross}}, \tag{42}$$

equal to £43,209 (S1) and £9,123 (S2). This is the same identity as E[L] in Section 7.3. The causal engine does not add a cancellation-loss term (Will_Cancel is hard-coded to 0). The dashboard's "annual dragons" is *int(|D| · 365/Δ)* — truncation, not rounding — giving 123 and 64 respectively (127 × 365/374 = 123.94, which *round* would send to 124).

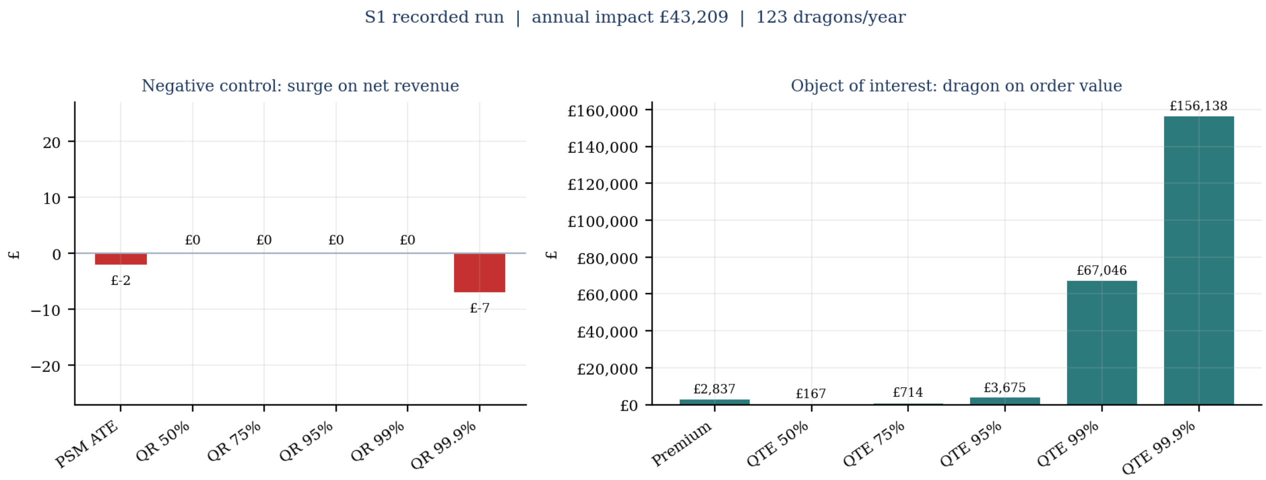

Figure 9. *Scenario 1, redrawn from Table 6 (the recorded run). Left: surge ATE/QR is the negative control (PSM −£2; QR 99.9% −£7). The dashboard PNG in results/ clipped the y-axis at 0 and labelled PSM as £7; that artefact is not used here. Right: dragon premium and QTEs, the object of interest.*

9. Hawkes self-exciting point processes

9.1 Intensity, compensator, branching ratio

Let $0 < t_1 < \dots < t_N \le T$ be the dragon timestamps, measured in minutes from the first dragon. A (univariate) Hawkes process with exponential kernel has conditional intensity

$$\lambda(t) = \mu_H + \sum_{t_i < t} \alpha_H e^{-\beta_H (t - t_i)}, \quad \mu_H > 0,\ \alpha_H \ge 0,\ \beta_H > 0. \tag{43}$$

Each event jumps the intensity by α_H and the jump decays at rate β_H. The process is a linear branching process in continuous time: immigrants arrive at rate μ_H, and each event produces offspring according to the kernel. The expected number of direct offspring of one event—the branching ratio—is

$$n = \int_0^\infty \alpha_H e^{-\beta_H s}\, ds = \frac{\alpha_H}{\beta_H}. \tag{44}$$

Stationarity in the Hawkes sense requires $n < 1$. If $n \ge 1$ the intensity explodes (subcritical vs critical branching). The half-life of a jump is $\log 2 / \beta_H$. These two numbers, not the raw α_H, are what “self-excitation confirmed” would have to mean.

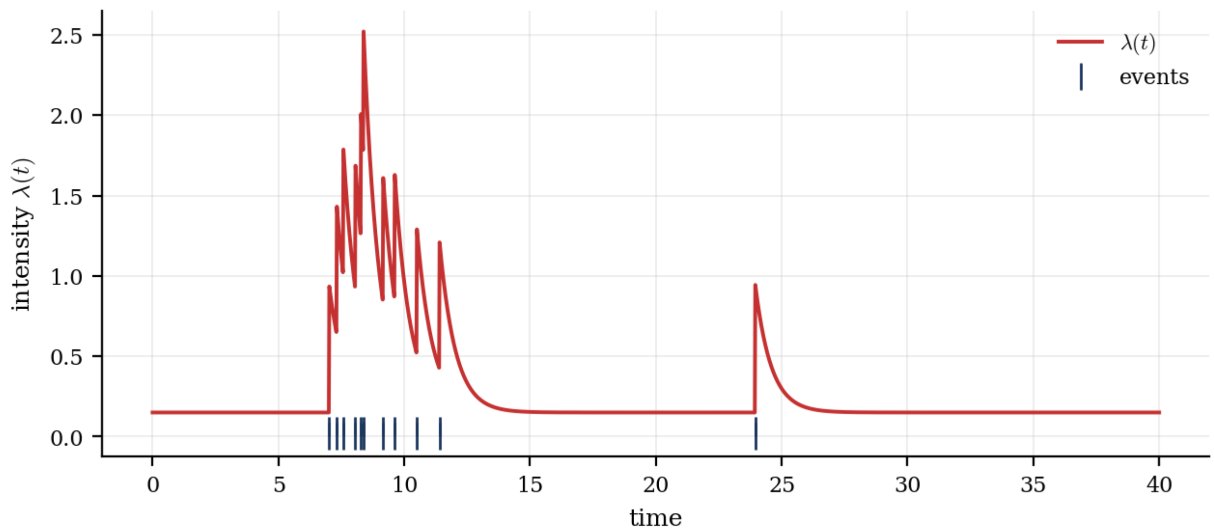

Figure 10. *A simulated exponential Hawkes intensity (not the warehouse fit): each event produces a jump that decays, and clustered events produce bursts. The warehouse MLE, by contrast, has α_H/β_H ≈ 0.002, so bursts die in a fraction of a second.*

9.2 Likelihood, from the point-process definition

For a simple point process on [0, T] observed as $t_1,\dots,t_N$, the log-likelihood (Daley and Vere-Jones; Ogata 1978) is

$$\ell(\mu_H,\alpha_H,\beta_H) \;=\; \sum_{i=1}^{N}\log\lambda(t_i) \;-\; \int_0^T \lambda(t)\,dt. \tag{45}$$

The integral is the compensator Λ(T) = E[N(T) | history], and for the exponential kernel it has the closed form

$$\int_0^T \lambda(t)\,dt \;=\; \mu_H T \;+\; \frac{\alpha_H}{\beta_H}\sum_{i=1}^{N}\bigl(1-e^{-\beta_H(T-t_i)}\bigr). \tag{46}$$

The sum over log $\lambda(t_i)$ is $O(N^2)$ if written naïvely. Ogata's recursion reduces it to O(N). Set $A_1 = 0$ and

$$\lambda(t_i) = \mu_H + A_i, \qquad A_{i+1} = (\alpha_H + A_i)e^{-\beta_H(t_{i+1}-t_i)}. \tag{47}$$

The negative log-likelihood is passed to L-BFGS-B with bounds $\mu_H \geq 10^{-8}$, α_H in [0, 0.99], $\beta_H \geq 0.1$, starting at $(2.5\times10^{-4}, 0.75, 1.2)$.

Implementation note — what the repository actually does.

Equation (47) is Ogata's recursion: $A_{i+1} = (\alpha_H + A_i) \exp(-\beta_H \Delta t)$, which decays the current jump on the way to the next event. The code is not that update. After recording $\log(\mu_H + A)$ it does *A = alpha + A * exp(-beta * dt)*, which adds an *undecayed* jump of size α_H at the next event. Those two recurrences agree only when $\Delta t = 0$. The missing decay is the material Hawkes bug: it makes almost every isolated event look as if it arrived on intensity $\approx \alpha_H$, so the MLE drives α_H to the box bound 0.99 and collapses μ_H to $\sim 10^{-6}$ per minute. Scenario 1's stationarity guard is additionally broken: *beta <= (alpha / beta) >= 1.0* is a chained comparison that is true only when $(\beta_H \le \alpha_H/\beta_H)$ and $(\alpha_H/\beta_H \ge 1)$. It does not implement $n < 1$. Scenario 2's guard is the correct $(\alpha_H/\beta_H) \ge 1$. The guard never binds at the recorded MLE ($n \approx 0.002$), so it is not why α_H sits on the bound. Both scenarios set $T = t_N$ (the last dragon), not the end of the 374-day sample, so the compensator omits the empty interval after the last event. Scenario 1 also updates A after the last event, using a stale inter-arrival; that update is never read. Scenario 2 builds timestamps from *Date* (midnight), not InvoiceDate, so same-day dragons have $\Delta t = 0$. The Scenario 2 print line hardcodes "Target annual: 70"; config.py has TARGET_ANNUAL_DRAGONS = 72. With $\beta_H \approx 410\ \text{min}^{-1}$ the branching ratio is $n \approx 0.990/410 \approx 0.0024$, and the half-life is $\log 2 / 410 \approx 0.1$ seconds. That n would describe almost-Poisson immigrants if μ_H were of order $N/T \approx 2{\times}10^{-4}$ per minute. The fitted μ_H is two orders of magnitude smaller, because the undecayed jump is soaking up the likelihood. Neither a Poisson process nor a genuine Hawkes burst is identified. The printed phrase "self-excitation confirmed" is not supported by n. A useful Hawkes fit on daily-aggregated dragons, or a kernel with a much smaller β_H, would be a different model.

9.3 Thirty-day expected counts

The 30-day forecast is the compensator increment from the last event,

$$\mathbb{E}[N(t_N+\Delta) - N(t_N) \mid \mathcal{H}_{t_N}] = \int_{t_N}^{t_N+\Delta} \lambda(s)\, ds, \tag{48}$$

with $\Delta = 30 \times 1440$ minutes, evaluated by the trapezoid rule on 15,000 grid points of a Numba-parallel intensity. The printed value is 1.5 dragons in both scenarios. Given $n \approx 0$ and $\mu_H \approx 1.9{\times}10^{-6}$ per minute, the immigrant contribution alone is $\mu_H\, \Delta \approx 0.08$. Residual excitation from the history is numerically zero after a few seconds. The gap between 0.08 and 1.5 is a quadrature/implementation residual (simultaneous timestamps, the intensity evaluated at $t = t_N$ including the last jump, trapezoid on a coarse grid relative to $\beta_H = 410$). It should not be read as a 30-day operational forecast.

Hawkes quantity	Scenario 1	Scenario 2
N (dragons)	127	66
μ_H (per minute)	1.9×10^{-6}	2.0×10^{-6}
α_H (jump)	0.990 (bound)	0.990 (bound)
β_H (per minute)	409.925	405.170
Branching ratio α_H/β_H	0.0024	0.0024
Half-life log 2 / β_H	≈ 0.10 s	≈ 0.10 s
Printed 30-day E[N]	1.5	1.5
Immigrant-only μ_H Δ	≈ 0.08	≈ 0.09

Table 7. *Fitted Hawkes parameters. The interesting number is the branching ratio, which is essentially zero. Self-excitation is not a feature of this fit.*

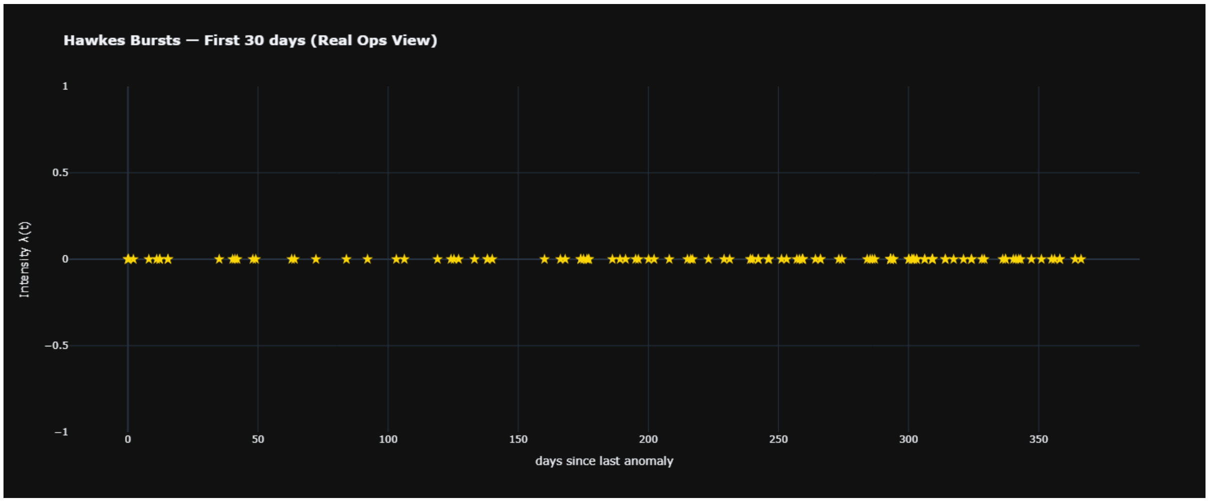

Figure 11. *Dashboard Hawkes plot, Scenario 1. The title says "first 30 days"; zoom_days = 367, so the x-axis is days from the first dragon. Intensity is evaluated only on [t_N, t_N+30 days] (t_N is the last dragon, near the end of the year), so the fitted λ is not in this window and does not appear. The gold markers are event times. With β_H ≈ 410 the half-life is ~0.1 s; this is not evidence of multi-day clustering.*

10. State-space projection of monthly dragon counts (Kalman, not Bayesian BSTS)

10.1 Naming

The repository folder is *hwk_bsts_forecasting* and older prose referred to PyMC / PyStan. Neither library is imported anywhere. What is implemented is a linear-Gaussian state-space model updated by the Kalman filter, with Q and R treated as fixed numbers. That is Harvey's structural time series in its filtering form, not a Bayesian structural time series (no posterior, no MCMC, no spike-and-slab). This paper calls it a Kalman local-linear-trend filter, which is what it is.

10.2 Local linear trend

Let y_t be the dragon count in month t, t = 1,…,m, with missing months filled by zero. A local linear trend is

$$y_t = \mu_t + \varepsilon_t, \quad \varepsilon_t \sim N(0, R) \tag{49}$$

$$\mu_t = \mu_{t-1} + \nu_{t-1} + \eta_t, \qquad \nu_t = \nu_{t-1} + \zeta_t \tag{50}$$

with process noises independent of ε, Q = diag(q_μ, q_ν) = diag(8×10^{-4}, 5×10^{-7}) on the trend block (the code also puts 8×10^{-6} on each unused seasonal coordinate), and R = numpy.var of the strictly positive y_t (1/n, not the unbiased 1/(n−1) sample variance). In state-space form, $x_t = (\mu_t, \nu_t)^T$,

$$x_t = Fx_{t-1} + w_t, \quad F = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad y_t = Hx_t + \varepsilon_t, \quad H = (1,\ 0). \tag{51}$$

F is the 2×2 local-linear-trend matrix with F_{11}=F_{12}=F_{22}=1 and F_{21}=0. That is exactly what the code realises by starting from an identity and setting F[0,1] = 1. The observation is y = x[0], i.e. H = (1, 0, …).

10.3 Kalman filter

Given a prior $x_{t|t-1}$, $P_{t|t-1}$, the prediction and update (Kalman, 1960) are

$$x_{t|t-1} = Fx_{t-1|t-1}, \qquad P_{t|t-1} = FP_{t-1|t-1}F^\top + Q \tag{52}$$

$$S_t = HP_{t|t-1}H^\top + R, \qquad K_t = P_{t|t-1}H^\top S_t^{-1} \tag{53}$$

$$x_{t|t} = x_{t|t-1} + K_t(y_t - Hx_{t|t-1}), \qquad P_{t|t} = P_{t|t-1} - K_t\,S_t\,K_t^\top \tag{54}$$

With H = (1, 0, …) one has $S_t = (P_{t|t-1})_{00} + R$ and $K_t = (P_{t|t-1}$ column 0)/S_t, which is the scalar form in the code. Initialisation is $x_0[0]$ = y-bar, P_0 = 50 I. The in-sample fit plotted in the dashboard is the sequence of one-step predictions H $x_{t|t-1}$.

10.4 The unused "seasonal" block

The code allocates a state of dimension 2 + 12 − 1 = 13, writes a shift-register into coordinates 2:13, and sets a −1 in the last row. A genuine dummy-seasonal model of period 12 would enter the observation as $y_t = \mu_t + \gamma_t + \varepsilon_t$, i.e. H would have a 1 on the first seasonal coordinate. The implemented H looks only at coordinate 0. The eleven seasonal coordinates are evolved by F and inflated by Q, and never observed. They do not affect the likelihood of y. The filter is a local linear trend with an inert companion. Calling it "local-level + seasonal BSTS" is, on the code as written, incorrect on three counts: it is not Bayesian, it is not local-level (there is a slope), and the seasonal is not in the measurement.

10.5 Projection, and the clip to {6,…,10}

Open-loop projection is $x_{m+k} = F^k\, x_{m|m}$, $y_{m+k} = (x_{m+k})_0$. The printed monthly rate is not the raw open-loop path. Let y-bar$_{10:12}$ be the mean of the last three projected months, and let *r* be its nearest integer (exactly *int(np.round(np.mean(proj[-3:])))* in *mle_bsts.py*). The code then projects that integer onto {6,…,10}:

$$\bar{y}_{10:12} = \tfrac{1}{3}(\hat{y}_{m+10} + \hat{y}_{m+11} + \hat{y}_{m+12}), \qquad \tilde{y} = \max(6, \min(r, 10)), \quad r = \mathrm{round}(\bar{y}_{10:12}). \tag{55}$$

then annualises as 12 y~. Scenario 1 hits the cap (10/month -> 120/year); Scenario 2 hits the floor (6/month -> 72/year). The "2012 forecast" is therefore a clipped integer in a five-point set, not a free Kalman forecast. TARGET_ANNUAL_DRAGONS in config.py is 104 (S1) and 72 (S2). The 72 equals the floor × 12 clip; 104 is not the cap of 120. The Monte Carlo module prints that target and does not use it in the path equation.

10.6 P&L bleed identity

Write N for the clipped annual count, $n_U = \Sigma\, U_i$, n_D = |D|, $U = \Sigma\, U_i\, V_i$. The anomaly premium used in this module is U / n_U (unfulfilled-dragon mean value), the breach rate is n_U / n_D, and

$$B \;=\; \hat{N} \cdot \tfrac{n_U}{n_D} \cdot \tfrac{U}{n_U} \cdot g \;=\; g \cdot \hat{N} \cdot \tfrac{U}{n_D}. \tag{56}$$

Scenario 1 hardcodes U = 147,583 and n_U = 66 rather than reading them from the frame; the identity still holds with N = 120, n_D = 127, g = 0.30, giving B = £41,835. Scenario 2 computes U and n_U from the frame and reports B = £10,197. Relative to the causal annual impact I = g (365/Δ) U, the bleed replaces the empirical annualisation 365/Δ · n_D by the clipped Kalman count N. That is the only structural difference.

10.7 Mitigation multipliers

Four scenario multipliers ρ in {0.70, 0.55, 0.15, 0.00} are applied to N, and savings are B(1−ρ). They are not estimated treatment effects of rerouting or safety stock. They are proportional-reduction arithmetic, labelled with operational names. They should be read as a sensitivity table on N, not as an operations-research optimisation of safety stock.

Quantity	Scenario 1	Scenario 2
Clipped monthly rate y~	10 (cap)	6 (floor)
Annual N = 12 y~	120	72
Bleed B = g N U / n_D	£41,835	£10,197
5-year arithmetic 5B	£209,173	£50,987
Reroute 30% (ρ=0.70), saving	£12,550	£3,059
Safety stock 7 SKUs (ρ=0.55), saving	£18,826	£4,589
Full mitigation (ρ=0.15), saving	£35,559	£8,668
Zero anomalies (ρ=0), saving	£41,835	£10,197

Table 8. *Kalman-clipped count forecasts and proportional mitigation arithmetic. Savings = B(1−ρ).*

Implementation note — what the repository actually does.
Scenario 1's P&L uses hardcoded 147583.0 and 66, which match the delay-stage unfulfilled totals on the recorded run but will silently desynchronise if the delay seed or dragon rate changes. Q and R are not estimated (no EM, no Gaussian MLE, no Bayes). With m ≈ 13 monthly points, estimating a 13-dimensional state would be unidentified anyway; fixing Q, R and clipping y~ is a computational convenience, not a statistical claim.

11. Peaks-over-threshold GPD on daily realised loss

Section 5 fitted a tail to *quantity*. The backtest module fits a GPD to a different random variable: daily realised loss. Define, on each ticket,

$$L_i^{\text{tk}} = U_i\,(V_i + H_i), \tag{57}$$

and aggregate $l_t = \Sigma_{i:\ \text{day}(i)=t} L_i^{\text{tk}}$. Most days have $l_t = 0$. The 99th percentile u of $\{l_t\}$ is the POT threshold (GPD_THRESHOLD_PCT = 99.0). Excesses $Y_j = l_t - u$ on $\{l_t > u\}$ are modelled as GPD(ξ, β_u), with MLE as in Section 5.10.

11.1 VaR and ES of the parent, from a GPD excess

Let n_{obs} be the number of distinct invoice dates in the daily-loss series (not 374 calendar days) and N_u the number of exceedances. For x > u the POT approximation is

$$P(\ell > x) \;\approx\; \frac{N_u}{n_{\text{obs}}}\Big(1 + \xi\frac{x-u}{\beta_u}\Big)^{-1/\xi}. \tag{58}$$

Value-at-risk at tail probability α (the code uses α = 0.05, i.e. a 95% VaR of the daily-loss distribution) solves P(l > VaR) = α, hence

$$\mathrm{VaR}_\alpha \;=\; u + \frac{\beta_u}{\xi}\left[\left(\frac{n_{\mathrm{obs}}\,\alpha}{N_u}\right)^{-\xi} - 1\right] \tag{59}$$

for $\xi \neq 0$, and $\mathrm{VaR}_\alpha = u - \beta_u \log(n_{obs}\,\alpha / N_u)$ for $\xi = 0$. The code also enforces $\mathrm{VaR}_\alpha \geq u$. That clamp is decisive here. The GPD is fitted above the 99th percentile, so $N_u/n_{obs} \approx 0.01$, while the code asks for a 95% VaR ($\alpha = 0.05$). The unconstrained 95% quantile lies *below* u, and the clamp therefore sets $\mathrm{VaR}_{0.95} = u$. Expected shortfall of a GPD parent has the closed form (McNeil, Frey and Embrechts)

$$\mathrm{ES}_\alpha \;=\; \frac{\mathrm{VaR}_\alpha + \beta_u - \xi u}{1-\xi}, \qquad \xi < 1, \tag{60}$$

which is the line *(var95_gpd + beta - xi * u) / (1 - xi)*. For ξ -> 0 this collapses to $\mathrm{VaR}_\alpha + \beta_u$, the exponential tail mean. Substituting the clamped $\mathrm{VaR}_{0.95} = u$ gives $\mathrm{ES}_{0.95} = u + \beta_u/(1-\xi)$, which is $E[\ell \mid \ell > u]$ — the GPD mean above the *99th* percentile, labelled as a 95% expected shortfall. Check: 3,990 + 30,440.76 / 0.952 ≈ 35,965, matching the printed £35,966.

11.2 Fitted numbers, and the N_u = 4 problem

GPD on daily realised loss	Scenario 1	Scenario 2
Threshold u (99th pctile of *l*_t)	£3,990	£1,745
Exceedances N_u	4	4
ξ	+0.0480	−0.0881
β_u	30,440.76	5,829.39
$\mathrm{VaR}_{0.95}$ (daily)	£3,990	£1,745
$\mathrm{ES}_{0.95}$ (daily)	£35,966	£7,102
Domain of daily loss	weakly Fréchet	Weibull (bounded)

Table 9. *POT GPD fitted to daily realised loss in the backtest module. These are daily, not annual, functionals. Printed VaR95 equals u because α = 0.05 lies below the 99th-percentile exceedance rate. The P&L print labels ES95 as "annualised"; the formula does not multiply by 252 or 365.*

Three comments. First, $N_u = 4$ in Scenario 1 is not a tail sample; it is four points. A two-parameter MLE on four excesses, with ξ bounded above by 1, will sit near the exponential (ξ ≈ 0), which is what 0.048 is. Scenario 2's P&L block never prints N_u; replaying the same seed and 99th-percentile rule gives $N_u = 4$ on $n_{obs} = 305$ invoice days. Second, Scenario 2's ξ = −0.088 is a Weibull-domain daily-loss tail: after netting, daily unfulfilled-dragon loss is a rare, bounded-looking compound, even though quantity itself is Fréchet. There is no contradiction. Quantity and daily loss are different variables, and the daily-loss series is mostly zeros. Third, the printed "$\mathrm{VaR}_{0.95}$ / $\mathrm{ES}_{0.95}$" pair is the 99th-percentile threshold and the GPD mean above it, because α = 0.05 sits below the exceedance rate and VaR is clamped to u. The printout's phrase "GPD ES95 (daily tail, annualised)" is a further labelling error: the number £35,966 is the daily ES, not multiplied by a year. Annualising it again inflates it by two orders of magnitude.

12. Purged expanding-window backtesting

12.1 The window algebra

Let the sorted unique dates be $d_1,\ldots,d_T$. An expanding window is indexed by i, starting at the first date at least MIN_TRAIN_DAYS = 90 after d_1, stepping by WINDOW_STEP = 7 dates, with a purge of PURGE_GAP = 3 days:

$$d^{\text{end}} = d_i, \quad d^{\text{cut}} = d^{\text{end}} - 3, \quad \text{train}:\ d \le d^{\text{cut}}, \quad \text{test}:\ d^{\text{cut}} < d \le d^{\text{end}}. \tag{61}$$

Windows with fewer than 50,000 training tickets or fewer than 5 test tickets are skipped. The purge exists so that the test slice does not share invoices, information, or overlapping delay draws with the end of train—the same reason López de Prado purges embargoed observations in combinatorial CV. On this file one obtains 34 scored windows in Scenario 1 and 33 in Scenario 2.

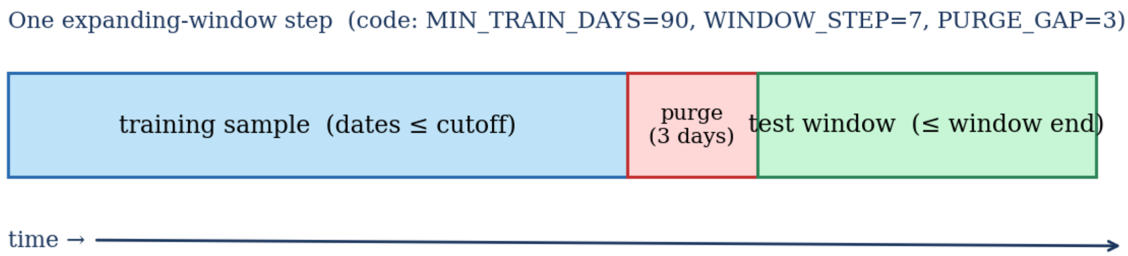

Figure 12. *One expanding-window step as implemented: train up to the cutoff, a three-day purge, a short test slice up to the window end. The violation test compares test realised loss to a window-matched Monte Carlo VaR, not to the annual VaR.*

12.2 Two Monte Carlo scales inside one loop

On each training slice the module re-estimates the log-normal of Section 7, with one crucial change: the mean daily dragon revenue is the mean over *dragon-active training days*, not over all calendar days. Paths of length 365 give an annual loss distribution; its 95% quantile and tail mean are the rolling annual VaR95/ES95 plotted in the dashboard. Independently, the first Δ_{test} columns of the same path matrix (Δ_{test} is the length of the test slice in days, typically 3) are summed to a window-matched loss; the 95% quantile of *that* short-horizon distribution is the VaR against which the test realised loss is scored. A violation is

$$I_k \ = \mathrm{I}\Big\{\ell_k^{\text{test}} > \widehat{\text{VaR}}_{0.95}^{(\text{window})}(k)\Big\}. \tag{62}$$

This scale-matching is the right idea: one must not test a three-day realised loss against an annual VaR. The loss definitions still differ. Test realised loss is $\Sigma\ (V_i+H_i)$ over unfulfilled dragons in the slice (gross, no margin). Window VaR is a quantile of $g\ r_V\ \Sigma\ R_t$ (margin-adjusted dragon revenue). A violation is therefore a comparison of two different currencies of loss. That does not make the test useless; it does mean a 0/34 violation rate is not a calibration theorem.

12.3 Kupiec's unconditional coverage test

Under a correctly calibrated VaR_{1-p} with $p = 0.05$, the indicators I_k are Bernoulli(p) (and, if the windows do not overlap in information, independent). Let n_w be the number of windows and k the number of violations. Kupiec's proportion-of-failures likelihood-ratio test of H_0: $E[I] = p$ is

$$\mathrm{LR_{uc}} \;=\; -2\log\frac{(1-p)^{n_w-k}p^k}{(1-\hat{\pi})^{n_w-k}\hat{\pi}^k} \;\to\; \chi^2_1, \tag{63}$$

with $\pi = k/n_w$. The p-value is $1 - F_{\chi^2\_1}(\mathrm{LR_{uc}})$.

Implementation note — what the repository actually does.
When $k = 0$ or $k = n_w$ the code returns p-value 1.0 and skips the ratio. That is the wrong number. For $k = 0$ the LR is $-2\, n_w \log(1-p)$, which for $n_w = 34$ and $p = 0.05$ equals 3.49, p-value ≈ 0.062, i.e. a borderline non-rejection, not a perfect 1.0000. The printout "Kupiec p=1.0000 | calibrated" overstates the evidence from a zero-violation string. Scenario 2, with 2/33 violations (6.1%), has a well-defined LR and Kupiec p = 0.7864, which is the more informative of the two tests.

12.4 Christoffersen's independence test

Independence of exceptions is tested as a first-order Markov chain against i.i.d. Bernoulli. Count transitions $n_{ij} = \#\{k : I_{k-1}=i, I_k=j\}$ for i,j in {0,1}, write $\pi_{01} = n_{01}/(n_{00}+n_{01})$, $\pi_{11} = n_{11}/(n_{10}+n_{11})$, $\pi = (n_{01}+n_{11})/n_{\bullet}$, and

$$\mathrm{LR_{ind}} \;=\; -2(\ell_{\mathrm{iid}} - \ell_{\mathrm{Markov}}) \;\to\; \chi^2_1. \tag{64}$$

The restricted likelihood is that of i.i.d. Bernoulli(π); the unrestricted is the two-probability Markov chain. Christoffersen's conditional coverage test (not computed here) would be $\mathrm{LR_{cc}} = \mathrm{LR_{uc}} + \mathrm{LR_{ind}} \sim \chi^2\_2$. Scenario 2 reports Christoffersen p = 0.6054: no evidence of clustered exceptions in a sample of 33 indicators, which is also a statement about power.

12.5 Last-window Monte Carlo, the other ES95

The P&L block of the backtest prints a second pair (VaR95_MC, ES95_MC) taken from the *last* expanding window's 365-day paths. Because that window uses the active-day mean, the numbers are much larger than Section 7: ES95_MC = £248,220 (S1) and £77,459 (S2). The older five-page report used these as headlines (~£268k / ~£77k). They are a different functional of a different mean. They are not a revision of £60,134 / £11,716.

Backtest quantity	Scenario 1	Scenario 2
Windows scored	34	33
Violations	0 (0.0%)	2 (6.1%)
Printed Kupiec p	1.0000 (see note)	0.7864
Corrected Kupiec p at k=0	≈ 0.062	—
Christoffersen p	1.0000	0.6054
Last-window MC VaR95 (annual)	£224,246	£72,267
Last-window MC ES95 (annual)	£248,220	£77,459
GPD daily ES95	£35,966	£7,102

Table 10. *Purged expanding-window backtest, as implemented (90 / 7 / 3), not the 180 / 30 / 14 of the outdated five-page PDF.*

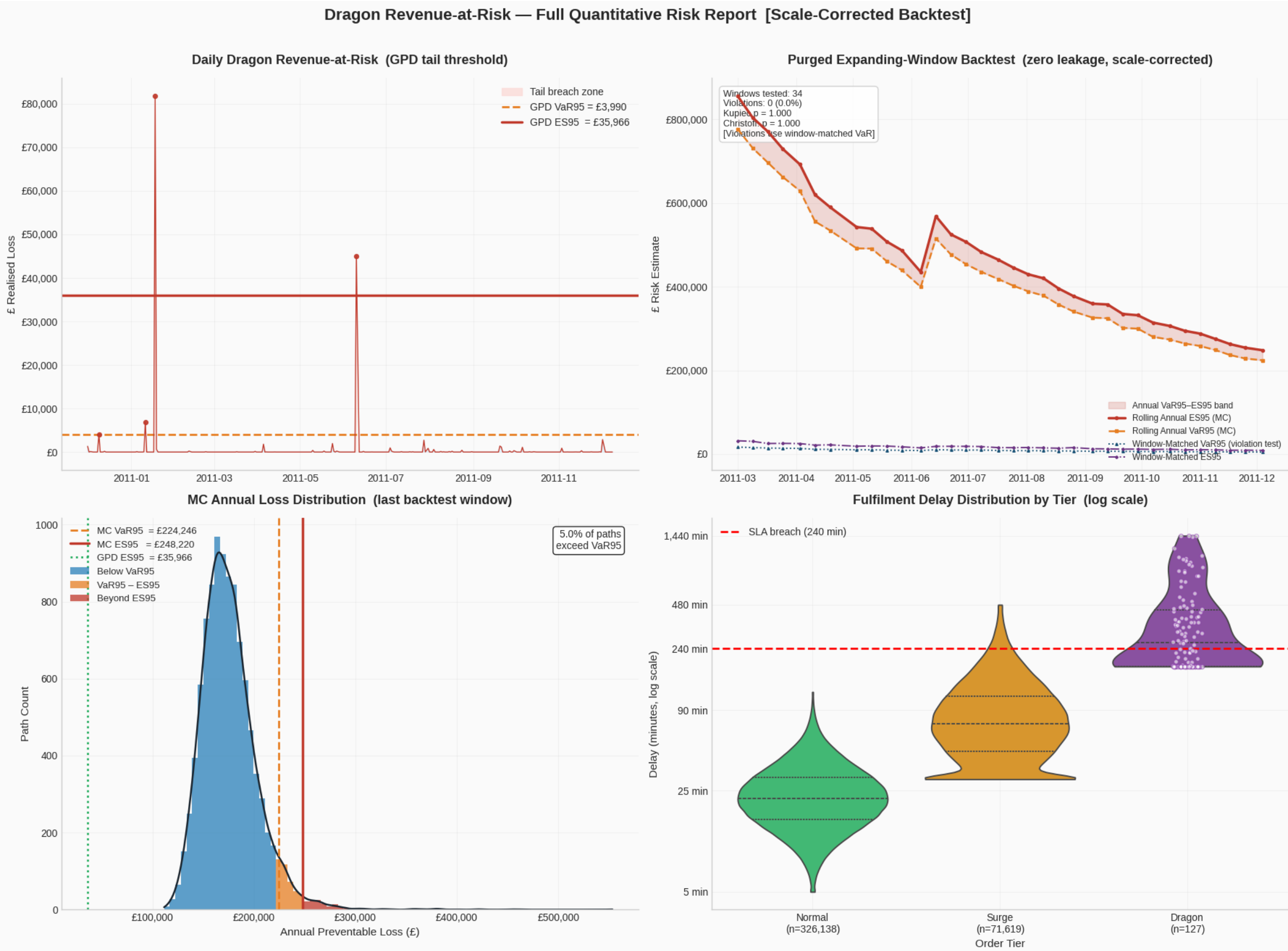

Figure 13. *Four-panel quantitative risk report, Scenario 1: daily loss with GPD thresholds; rolling annual and window-matched VaR/ES; last-window MC histogram; log-delay violins by tier.*

Implementation note — what the repository actually does.

The backtest re-draws delay tiers with the same seed, rather than consuming the delay-stage frame. It is a replay, not a continuation. MIN_TRAIN_DAYS=90, WINDOW_STEP=7, PURGE_GAP=3 live in config.py; any document quoting 180 / 30 / 14 is describing a previous version. Expected violations under p=0.05 are about 1.7 in 34 windows: zero is not surprising, and not a validation win. Heavier tails plus a higher VaR make zeros easier. Scenario 1 computes *date_range_days* as (max−min).days without +1 (373 rather than 374), so the backtest's annualised gross prints as £8,720,279 rather than (365/374) Σ V. Scenario 2 includes +1. The delay-stage unfulfilled flag also requires the SLA label to be Breach or Failure; the backtest uses only Delay_min ≥ τ. On these bins the two predicates coincide.

13. Reconciliation of every reported sterling figure

The repository prints several loss numbers per scenario. They look like rivals. They are not. Each is a different functional, of a different random variable, sometimes with and sometimes without the 30% margin, sometimes using all calendar days and sometimes only dragon-active days, sometimes annual and sometimes daily. This section writes the identity for each number, plugs in the recorded run, and states what question that number actually answers. Each identity is written so the chain can be audited from the text, without opening the code.

13.1 The primitives, Scenario 1

On the recorded gross run the following primitives are fixed by the delay stage (seed 314159):

- n = 397,884 tickets; Δ = 374 days; ΣV_i = £8,911,407.90.
- $|D|$ = 127 dragons; $\Sigma_D V_i$ = (implied by the rest); $\Sigma U_i V_i = U$ = £147,582.95; n_U = 66.
- g = 0.30; h = 0.25; ΣH_i = £75,635.58.
- Broader SLA-breach revenue (all tiers) = £694,291.

13.2 The identities

(A) Holding drag, not a tail functional. $\Sigma H_i = \Sigma V_i (D_i/1440)\, h$ = £75,635.58. Question answered: how much working-capital cost does the simulated delay clock charge, on every ticket.

(B) Broad SLA revenue-at-risk. $\Sigma \{V_i : SLA_i$ in {breach, failure}$\}$ = £694,291, annualised ×365/Δ to £677,583. Question answered: value sitting on any SLA miss, dragons or not. Not used downstream.

(C) Unfulfilled-dragon revenue, sample. $U = \Sigma U_i V_i$ = £147,583. Question answered: value sitting on dragons that cleared the 4-hour threshold in this 374-day file.

(D) Data-derived $\mathrm{EL}_{\mathrm{gross}}$.

$$\mathrm{EL}_{\mathrm{gross}} = \frac{365}{\Delta} U = \frac{365}{374} \times 147{,}583 = 144{,}031. \tag{65}$$

Question answered: (C), stretched to a 365-day year, still gross of margin. This is the Monte Carlo module's "EL".

(E) Margin-adjusted annual impact, also E[L] of the i.i.d. MC.

$$I = g \cdot \mathrm{EL}_{\mathrm{gross}} = 0.30 \times 144{,}031 = 43{,}209. \tag{66}$$

Question answered: (D) with a 30% gross-margin factor. This is the causal dashboard's "total annual impact", and it is the mean of the Monte Carlo loss L of Section 7, up to Monte Carlo error and the CV-cap.

(F) Monte Carlo median, VaR, ES of L. $L^{(p)} = g\, r_V \Sigma_{t=1}^{365} R_t^{(p)}$ with R_t log-normal of all-day mean and active-day CV. Median £42,387, VaR95 £54,579, VaR99 £62,998, ES95 £60,134. Question answered: the distribution of one-year margin-adjusted preventable loss under i.i.d. daily dragon revenue. The headline of *monte_carlo.py*.

(G) Kalman/Hawkes bleed.

$$B = g \cdot \hat{N} \cdot \frac{U}{n_D} = 0.30 \times 120 \times \frac{147{,}583}{127} = 41{,}835. \tag{67}$$

Question answered: (E), but with the printed annual count int(|D|·365/Δ) = 123 (the raw product is 123.94) replaced by the clipped Kalman count 120. Hardcoded U and n_U in Scenario 1.

(H) Last-window backtest MC ES95. Same path equation as (F), but E[R] is the mean of dragon-active days in the last training slice, not V_d over all calendar days, and the training slice is not the full sample. ES95 = £248,220 (S1), £77,459 (S2). Question answered: a more aggressive annual loss, conditional on "a dragon day looks like a dragon day". This is the number the outdated five-page report led with.

(I) GPD daily ES95. Printed as ES of the daily realised-loss series under a POT GPD at $\alpha = 0.05$: £35,966 (S1), £7,102 (S2). Because the GPD is fitted above the 99th percentile and VaR is clamped to u, this is $E[\ell \mid \ell > u_{99}]$ (V+H, no margin), not a 95% expected shortfall. Not annualised, despite the print label.

(J) "Tail scenario daily × 252". 99th percentile of daily realised loss, times 252: £1,005,366 (S1), £439,753 (S2). Question answered: a crude annualisation of a bad day as if it occurred every trading day. It is an upper-stress exhibit, not an expectation, and it double-counts the rarity of dragons. It should not be added to (F) or (H).

(K) Margin impact in the backtest P&L. g times (H): £74,466 (S1), £23,238 (S2). Question answered: (H) with an extra margin factor. Because (H) is already margin-adjusted in the path equation (× g inside annual_loss_mc), this is a *second* multiplication by 0.30, i.e. g^2 times a gross-style annual MC. It is the one number in the printout that is internally inconsistent with Section 7's convention.

Remark.
Row (K) is the only arithmetic error, as opposed to a labelling or functional-difference issue. annual_loss_mc already contains GROSS_MARGIN, and the P&L then prints es95_mc * GROSS_MARGIN again. A corrected "margin impact" in the backtest block would simply be (H) itself. The paper leaves (K) in the table so the double count is visible rather than silently dropped.

13.3 Master reconciliation table

ID	What is printed	S1	S2	Already ×g?	Horizon
A	Holding cost Σ H	£75,636	£59,274	n/a	sample
B	Broad SLA RaR	£694,291	£463,771	no	sample
C	Unfulfilled-dragon U	£147,583	£31,159	no	sample
D	EL_gross = (365/Δ)U	£144,031	£30,409	no	year
E	Causal annual impact g(365/Δ)U	£43,209	£9,123	yes	year
F	MC module ES95 of L	£60,134	£11,716	yes	year
F'	MC module VaR95 of L	£54,579	£10,989	yes	year
G	Kalman bleed g N U / n_D	£41,835	£10,197	yes	year
H	Last-window MC ES95	£248,220	£77,459	yes	year
H'	Last-window MC VaR95	£224,246	£72,267	yes	year
I	GPD ES of daily l_t (clamped; ≈ ES_{99})	£35,966	£7,102	no (V+H)	day
J	p99(l_t) × 252	£1,005,366	£439,753	no	pseudo-year
K	g × (H) [double count]	£74,466	£23,238	twice	year

Table 11. *Master reconciliation. (E), (F) and (G) are the coherent family of one-year, margin-adjusted, full-sample functionals. (H) is the same family with an active-day mean and a last-window train slice. (I) is a daily POT mean above the 99th-percentile threshold, printed as ES95 because VaR is clamped to u. (J) is a stress exhibit. (K) double-applies the margin.*

13.4 Which number should a reader remember

If the question is "in a bad year, under the i.i.d. log-normal of Section 7, what is preventable dragon loss at the 95% expected-shortfall point, margin-adjusted, on the full sample?", the answer is (F): about £60k gross view, about £12k netted view. If the question is "what is the expected margin-adjusted annual unfulfilled-dragon revenue, no tail, just a stretch of this year?", the answer is (E): about £43k / £9k. If the question is "what does the last expanding window's more aggressive MC say?", the answer is (H): about £248k / £77k. None of these is a forecast of a real warehouse. They are three well-defined functionals of a simulated file, and they differ for reasons that are now algebraic rather than mysterious.

14. Data governance and WMS implications

14.1 The 80,995 / −80,995 pattern

StockCode 23843, PAPER CRAFT LITTLE BIRDIE, enters the gross file as Q = 80,995, V = £168,469.60, the sample maximum, and is followed on a short clock (on the order of twelve minutes in the raw file) by a negative-quantity row of the same magnitude. It is the canonical UCI outlier. INFERENCE.md records two interpretations, and both are live.

Interpretation 1 — data-entry or integration error. A false large positive is booked and then reversed. The gross file then contains a phantom dragon. Any risk number that uses Scenario 1 without a matching rule will treat a keystroke as a £168k exposure. In a production WMS this is a data-quality incident: it should be caught by an order-entry validator (quantity caps, velocity checks, confirmation on outliers) long before it reaches a tail-risk engine. Scenario 2's netting is a blunt annual analogue of that validator.

Interpretation 2 — genuine booked demand, subsequently cancelled. The warehouse may already have allocated inventory, reserved labour, or purchased inbound by the time the negative row arrives. Gross exposure is then the economically right object for the hours between book and cancel, and netting after the fact understates the operational tail. Working capital was committed against £168k, not against zero. A risk system that only ever sees netted tickets will never provision for that window.

The two-scenario design is the paper's way of not choosing. A WMS with an audit trail would choose: it would timestamp allocation, pick, pack, ship and cancel, and the unfulfilled-dragon indicator of Section 6 would be an observed random variable rather than a simulated one. Until that file exists, reporting both the phantom-sensitive gross view and the netted view is the honest move.

14.2 A WMS KPI layer this project does not have

INFERENCE.md lists the standard execution KPIs that would close the loop between a dragon event and operations: order cycle time, perfect-order rate, fill rate, dock-to-stock time, inventory accuracy, pick accuracy, SLA attainment. Each of those is a time-stamped, system-of-record measurement. None of them is in UCI Online Retail. Attaching them to the dashboard is a product idea, not a result. The mathematical pipeline in this paper is the risk half of that product; the KPI half requires proprietary telemetry the author does not have.

14.3 What would change in a live WMS

- Dragon would be an observed predicate (value and cycle-time jointly exceeding a policy threshold), not a weighted sample.
- Delay D_i would be pick-complete minus order-entry, not a clipped log-normal.

- The surge flag would be an observed capacity state (wave, labour, carrier), and the ATE of surge on cycle time would become a genuine causal question, identified by whatever operational instrument is available (shift changes, weather, promotions).
- QTE of dragon on value would still be of interest, but it would be a description of a policy-defined set rather than a restatement of the sampling weights.
- Hawkes would be fitted to observed event times with a kernel whose time unit matched the clustering scale (hours or days, not a 0.1-second half-life).
- Backtests would run daily, on rolling proprietary history, with enough exceptions for Kupiec to have power.

15. Limitations, and what a production version would change

This section records the limits of the work, not an apology. Every item was accepted at the outset as a consequence of public data, undergraduate compute, and a one-year retail file.

15.1 Structural

- Delays are synthetic. No result about SLA, unfulfilment, holding, MC, Hawkes timestamps, or backtest realised loss is an observation of a warehouse.
- Dragon labels are assigned as a function of value. QTEs are partly mechanical. ATEs of surge are a negative control.
- Data vintage is 2010–2011. Fulfilment technology, carrier networks and customer expectations have moved.
- GPD and Hawkes are fitted on the full sample, not purged. EVT standard errors ignore the daily dependence of Section 4.3.
- One year of tickets gives 34 backtest windows. Coverage tests have low power. Zero violations in Scenario 1 is not a validation of the VaR.
- Five-year arithmetic (5B, 5 × ES) assumes a constant dragon rate, margin and SLA policy.

15.2 Statistical shortcuts, as implemented

- Monte Carlo paths are i.i.d. log-normal; Hawkes clustering is not injected (no Ogata thinning of the annual paths).
- CV is capped at 3; mean and CV of daily revenue are taken on different day-sets in the MC module versus the backtest module.
- PSM has no balance diagnostics and no double robustness; the propensity logit is fit on 50,000 rows (a 250,000-row fit does not complete); QR is a single subsample, no clustered standard errors, no bootstrap in the current config (BOOTSTRAP_REPS is unused). Scenario 1 matching reuses controls; Scenario 2 does not.
- Country is dropped by the loader; Hour collapses in Scenario 2. Propensity covariates are degenerate. Scenario 2's PSM/QR outcome is order value, not net of holding.
- Kalman Q, R are fixed; the seasonal block is unused; the monthly forecast is clipped to {6,…,10}. TARGET_ANNUAL_DRAGONS is printed in Monte Carlo and unused in the path equation.
- Hawkes Ogata recursion in the code is $A \leftarrow \alpha + A\, e^{-\beta\, \Delta t}$, not $(\alpha + A)\, e^{-\beta\, \Delta t}$: the current jump is not decayed. That is why α_H sits on the 0.99 bound and μ_H collapses. Branching ratio ≈ 0.002; T is the last event, not the sample end; Scenario 1's stationarity guard is an operator-precedence bug that does not bind at the MLE.

Scenario 2 timestamps are calendar dates (midnight). The S2 print hardcodes “Target annual: 70” against config 72.

• GPD daily MLE uses four exceedances in each scenario (S2’s P&L does not print N_u; a same-seed replay gives 4 on 305 invoice days). The printed “VaR95/ES95” pair is the 99th-percentile threshold and the GPD mean above it, because $\alpha = 0.05$ lies below the exceedance rate and VaR is clamped to u. Quantity is never given a GPD MLE; AMSE Hill is an excess-Hill with $k = N_u$, and u* is unused. Shapiro–Wilk uses an unseeded 5,000-row subsample. Causal “annual dragons” is int(·), not round(·).

• Kupiec p-value is returned as 1 when k = 0.

• Backtest P&L multiplies an already-margined ES by g a second time. Scenario 1 annualises gross with 373 days rather than 374.

• Scenario 1 Hawkes P&L hardcodes 147583 and 66.

15.3 What I would do next, given a WMS extract

Integrate Hawkes intensity into the Monte Carlo by thinning: draw each day’s dragon count from a time-changed Poisson rather than from an i.i.d. log-normal revenue. Re-fit GPD and Hawkes inside the purged window, not on the whole sample. Estimate Q, R by EM or by a proper Bayesian structural time series if a posterior is actually wanted. Put Country and a genuine hour-of-day back into the propensity model, and add a love plot. Define dragon from observed cycle time × value, so that QTE is a contrast of a policy set rather than of a weighted sample. Replace the four mitigation multipliers by a small operations model (safety stock on the seven SKUs that actually dominate U). None of that is in this repository, and claiming it is would be the wrong kind of completeness.

16. Conclusion

The warehouse tail-risk problem, as posed here, is a problem in extreme value theory with a risk-management translation layer. Quantity on high-volume SKUs is in the Fréchet domain; three tail-index estimators and a positive mean-excess slope agree on that. Because the public file has no fulfilment clock, delays and dragon labels are assigned, and the only honest causal statement is: the surge ATE on net revenue is a negative control (empirically −£2 gross, −£18 netted), and the QTE of the dragon grouping on order value is the descriptive object of interest (£167 at the median, £3.7k at the 95th, £67k at the 99th, gross view). Annual preventable loss is then a Monte Carlo functional of dragon revenue, an empirical breach rate, and a 30% margin; its full-sample $ES_{0.95}$ is about £60k (gross) and £12k (netted). A Kalman local-linear-trend filter, not a Bayesian BSTS, projects a clipped monthly count; a Hawkes MLE on minute-scale timestamps sits on an α-bound and implies essentially no self-excitation. A purged expanding-window backtest, with the coverage tests written out, is the right validation idea applied to a sample too short to validate much.

The paper’s usable contribution is not a sterling number to take to a CFO. It is a chain of identities—from regularly varying tails, through an assigned delay, to a family of loss functionals that have been written down and reconciled—and a clear list of where the code and the theory part company.

What this monograph demonstrates, and what it does not.
It demonstrates that the full mathematical stack (EVT on quantity, POT/GPD on daily realised loss, log-normal MC, QTE, Hawkes MLE, Kalman filtering, Kupiec/Christoffersen) can be derived and wired, end to end, on a public retail file, with every sterling headline identified as a named functional. It does not demonstrate a validated operational model, a real SLA effect, a genuine Hawkes burst on a warehouse clock, or a policy prescription to hold 99th-percentile service on dragons. Those claims would require observed delays.

Appendix A. Notation, configuration, and additional derivations

A.1 Notation

Symbol	Meaning
V_i, Q_i, P_i	order value, quantity, unit price of ticket i
T_i, D_i	assigned delay tier; delay in minutes
Δ_i, W_i, U_i	dragon flag; surge flag; unfulfilled-dragon flag
H_i, R_i^{net}	holding cost; net revenue $V - H$
ξ, u, β_u	EVT index; threshold; GPD scale
μ_H, α_H, β_H, n	Hawkes baseline, jump, decay, branching ratio α_H/β_H
μ_l, σ_l	log-normal location and scale of delay
μ_d, σ_d	log-normal location and scale of daily dragon revenue
h, g	annual holding rate 0.25; gross margin 0.30
L, l_t	annual MC loss (margin-adjusted); daily realised loss (V+H)
$\delta(q)$, π	QTE of dragon on V at quantile q; dragon mean premium
I, B	causal annual impact $g(365/\Delta)U$; Kalman bleed $g\ N\ U\ /\ n_D$

Table 12. *Notation used in the body. Subscripts H, l, d, u distinguish Hawkes, delay-log-normal, daily-revenue-log-normal, and GPD-scale parameters that would otherwise share letters.*

A.2 Scenario configuration, as in config.py

Parameter	Scenario 1	Scenario 2
SURGE_PCT	0.18	0.18
DRAGON_PCT	0.00032	0.00025
DRAGON_BIAS_EXP γ	1.35	1.20
NORMAL / SURGE / DRAGON mean (min)	25 / 90 / 420	25 / 90 / 360
σ_*l* N / S / D	0.50 / 0.65 / 0.85	0.50 / 0.65 / 0.82
Clip N / S / D (min)	[5,120] / [30,480] / [180,1440]	[5,120] / [30,360] / [200,1440]
SLA bins (min)	[0, 30, 120, 480, ∞)	[0, 30, 120, 360, ∞)
SLA_BREACH_MIN τ	240 min (4 h)	360 min (6 h)
ANNUAL_HOLDING_RATE *h*	0.25	0.25
GROSS_MARGIN g	0.30	0.30
TARGET_ANNUAL_DRAGONS	104	72
MIN_TRAIN_DAYS / STEP / PURGE	90 / 7 / 3	90 / 7 / 3
PSM caliper / PS subsample	0.005 / 50,000	0.005 / 50,000
QR subsample	180,000	180,000
QUANTILES in current config.py	[0.95, 0.999]	[0.95, 0.999]
QUANTILES on the recorded run (Table 6)	{0.5, 0.75, 0.95, 0.99, 0.999}	same
RNG seed	314159	314159

Table 13. *All of these are assumptions, not estimates from warehouse telemetry. WERC/CSCMP public summaries inform the delay means, holding rate and SLA windows. Dragon rates (3.2 bps / 2.5 bps) and bias exponents put a thin, expensive tail in the dragon bucket; they are not a 99th-percentile cutoff on value. Table 6's QR/QTE rows use the recorded-run quantile list; current config.py keeps only [0.95, 0.999]. The propensity logit is fit on 50,000 rows (Section 8.3): a 250,000-row fit does not return on this file.*

A.3 Score equations of the GPD

For completeness, the GPD scores used by any gradient-based MLE (the backtest uses L-BFGS-B on $-l$) are, writing $z_j = Y_j / \beta_u$ and $\tau_j = 1 + \xi z_j$,

$$\frac{\partial \ell}{\partial \xi} = \frac{1}{\xi^2}\sum_j \log \tau_j \;-\; \left(1+\frac{1}{\xi}\right)\sum_j \frac{z_j}{\tau_j}, \tag{68}$$

$$\frac{\partial \ell}{\partial \beta_u} = -\frac{N_u}{\beta_u} \;+\; \left(1+\frac{1}{\xi}\right)\sum_j \frac{\xi Y_j}{\beta_u^2 \tau_j}. \tag{69}$$

At $\xi = 0$ the exponential scores take over by continuity. The code *rejects* $\xi \geq 1$ (barrier plus box bound) because $E[Y] = \beta_u/(1-\xi)$ is then infinite, without which ES is undefined. That identity is also why the mean-excess function of an exact GPD is $\beta_u/(1-\xi)$, not $(\beta_u + \xi u)/(1-\xi)$.

A.4 Hill variance and a caution on sqrt(k)

In the exact Pareto model, $\mathrm{Var}(\xi^{\mathrm{Hill}}) \approx \xi^2/k$. With k = floor(sqrt(n)) = 630 and $\xi \approx 0.5$ this is a standard error of about 0.02, which would make 0.515 look precise. The model is not exact Pareto, the sample is dependent, and k = sqrt(n) is a rule of thumb (not Hall's AMSE-optimal k); n is the transaction count, not the Pareto-filtered length. The paper therefore never quotes a confidence interval around Hill on this file. The AMSE curve of Section 5.11

is a stability diagnostic of a different Hill-type statistic (excess-Hill, $k = N_u$), not a standard error for (17), and u* is not used to fit a GPD on quantity.

A.5 Why last-window ES exceeds full-sample ES

Let f be the fraction of calendar days that are dragon-active. The MC module uses $E[R] = f \cdot \mu_{active}$; the backtest uses $E[R] = \mu_{active}$. The annual sum then scales by 1/f. With 127 dragons in 374 days, if each dragon-day carried one dragon, $f \approx 0.34$ and $1/f \approx 3$. A ratio of 248,220 / 60,134 ≈ 4.1 is in that neighbourhood, with the residual explained by a different training slice, a different r_V, and the CV cap. The ratio is a mean-definition effect, not a discovery that the last window was four times riskier.

A.6 Seed

Every numpy generator in the delay, Monte Carlo and backtest stages is constructed as default_rng(314159). Two other draws are *not* on that generator: the Shapiro–Wilk subsample of Section 4.4, and the AMSE bootstrap of Section 5.11, both of which call np.random.choice on the global NumPy RNG. The recorded run is therefore a single point in the seed space for the loss functionals, and a slightly moving point for those two diagnostics. A production stress test would report a distribution over seeds; this monograph reports the number the repository actually emitted.

A.7 Additional empirical exhibits

The following figures are produced by the pipeline (or, for the Scenario 2 causal panel, redrawn from Table 6) so that the mathematical claims in the body can be checked against the recorded run.

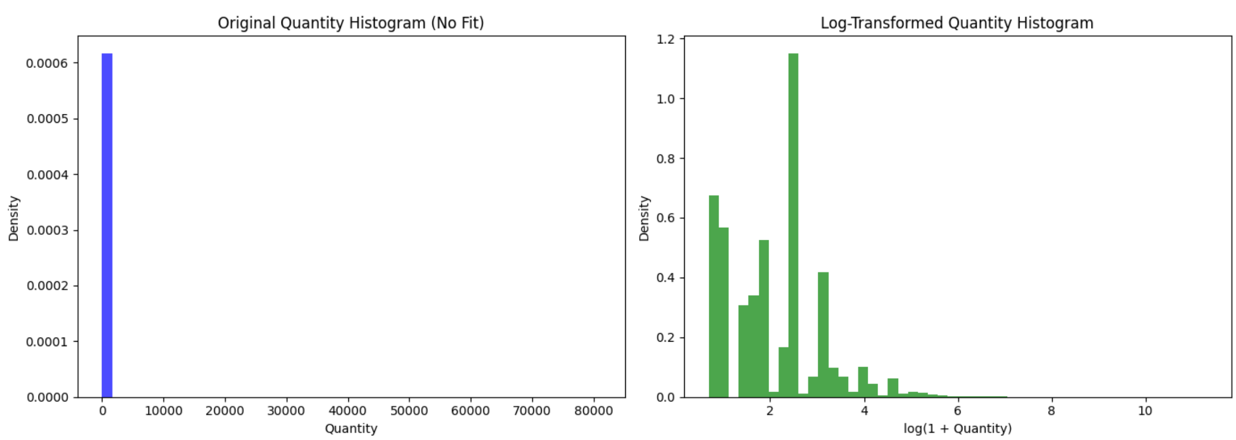

Figure 14. *Quantity on the original and log(1+Q) scales, Scenario 1, Pareto-filtered sample. The left panel is unusable as a density; the right panel is still right-skewed.*

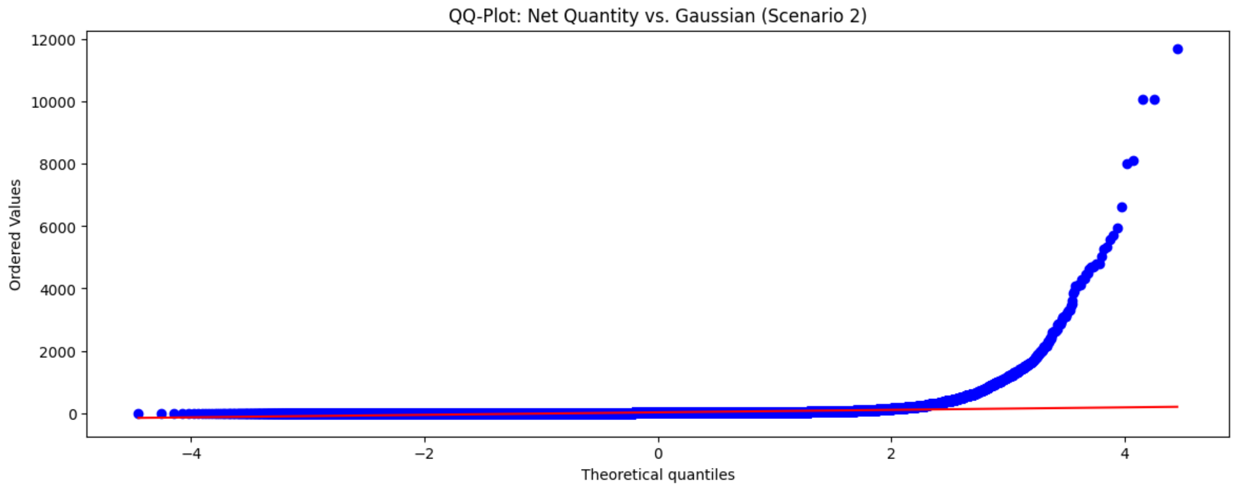

Figure 15. *Normal QQ-plot of net quantity, Scenario 2. The right-tail bend survives netting.*

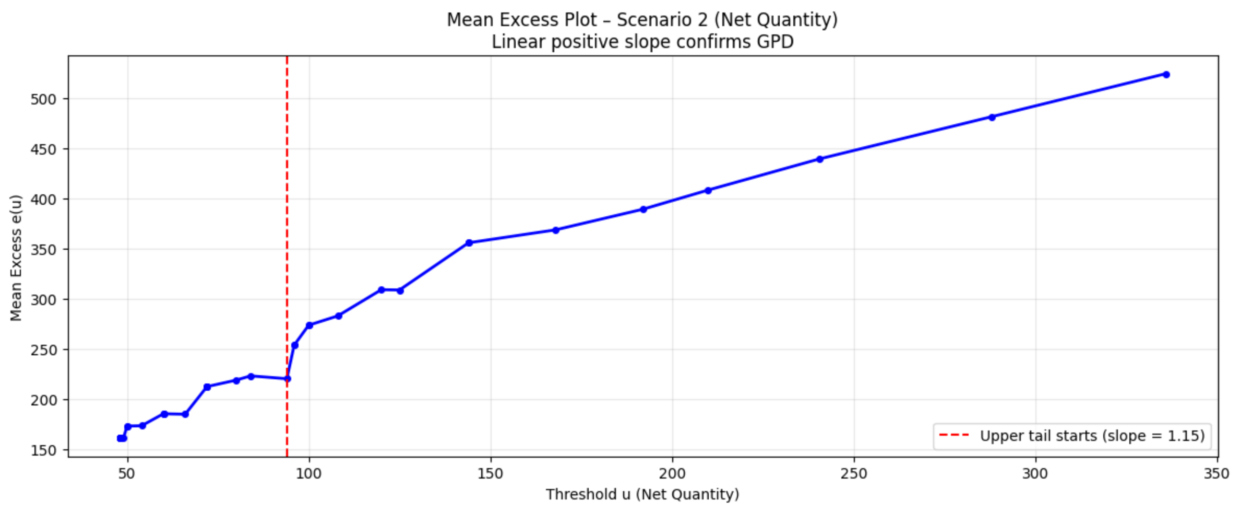

Figure 16. *Mean-excess plot, Scenario 2. Table 3's recorded-run upper-tail slope is +1.089; the PNG legend reads 1.15 (a later artefact). The qualitative message is unchanged: the slope is positive.*

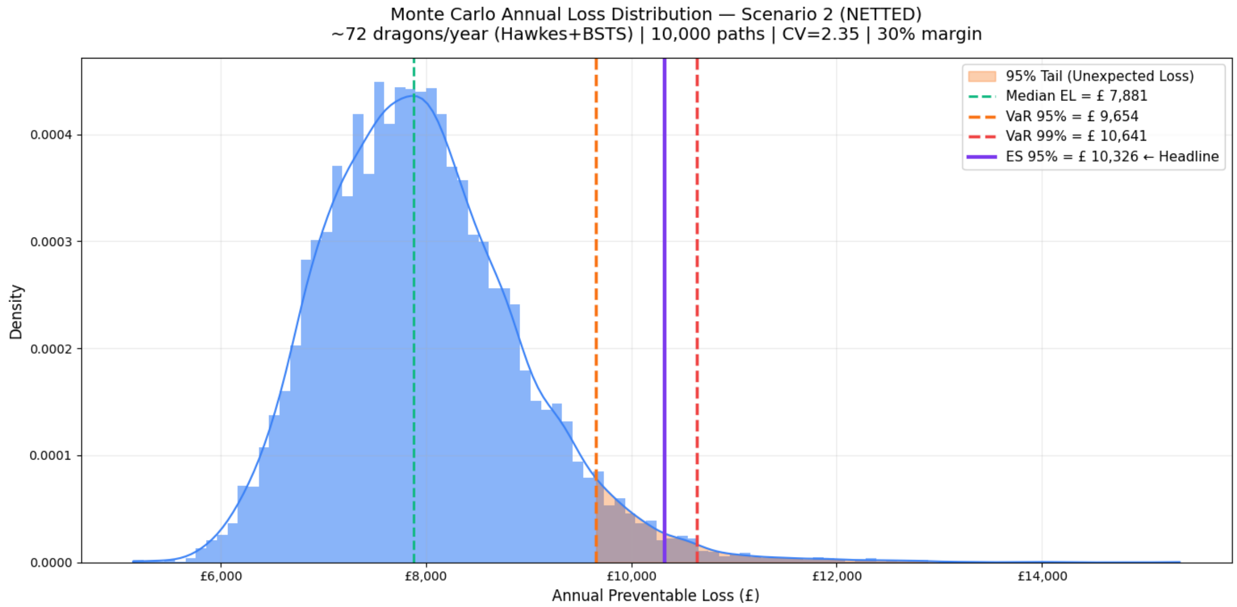

Figure 17. Monte Carlo annual loss density, Scenario 2. Same construction as Figure 8, smaller scale after netting. The PNG in results/ is a later artefact (legend: median £7,881, ES95 £10,326, CV=2.35). Table 5's recorded run is median £9,021, ES95 £11,716, CV=2.28; use the table, not the legend.

Figure 18. Four-panel risk report, Scenario 2 (2/33 violations). This is the more informative coverage test of the two scenarios.

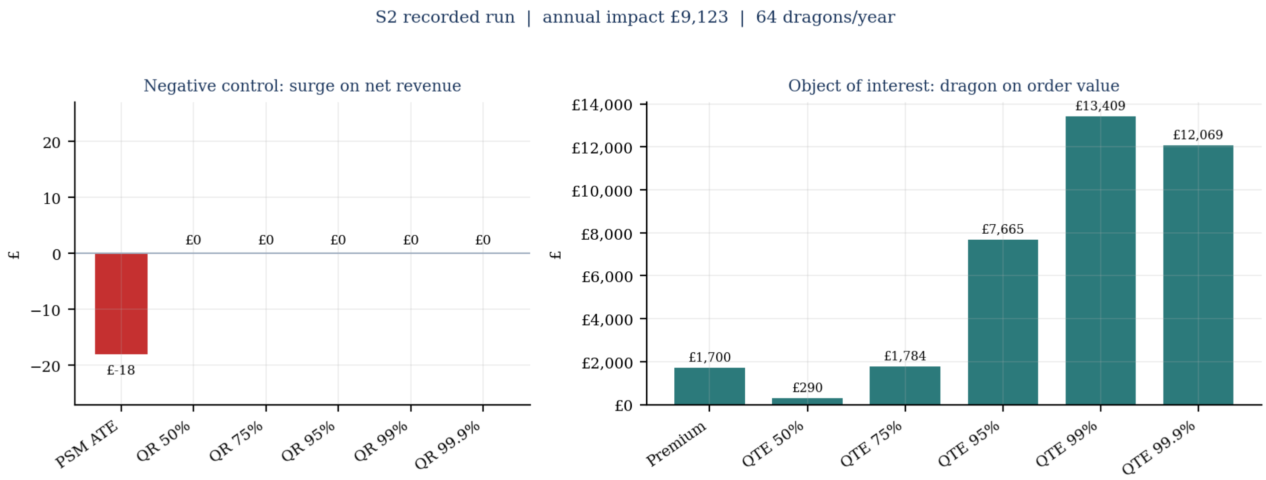

Figure 19. *Scenario 2, redrawn from Table 6 (the recorded run). Left: surge ATE is the negative control (PSM −£18; QR coefficients £0). The dashboard PNG in results/ was a later artefact (£7,977 annual impact, premium £1,609) and is not used here. Right: dragon premium and QTEs after netting; the 99.9th QTE sits below the 99th (66 dragons).*

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